

PREPARING SKILLS.
BUILDING CAREERS.
CREATING IMPACT.

DATA SCIENTIST *Career Guide* AND INTERVIEW PREPARATION

My Learnings, Insights & Practical Notes
for Aspiring Data Scientists



RESUME &
COVER LETTER



PORTFOLIO
DEVELOPMENT



INTERVIEW
PREPARATION



COMMUNICATION
& PRESENTATION



NEGOTIATION &
PROFESSIONAL
GROWTH

“

Technical skills help you get noticed,
but **professional preparation** helps you get **selected**.

”



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Data Scientist | AI Engineer



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Learning. Building. Sharing. Growing.

- Career Guidance
- Job Search Strategy
- Interview Success
- Professional Growth
- Making an Impact

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Module 1: Building a Foundation

1 Understanding the Role of a Data Scientist

1.1 Learning Objectives

After studying this topic, we will be able to:

- Summarize the role and tasks of data scientists.
- Identify some of the main skills a data scientist needs.
- Describe some possible career paths for data scientists.

1.2 Roles in the Data Ecosystem

In the industry, opinions differ about the skills and responsibilities of different roles in the data ecosystem. Moreover, many job titles and descriptions in data science appear to have overlapping responsibilities.

For comparison, the following are some of the major roles in the data ecosystem.

1.2.1 Data Analyst

Data analysts translate data and numbers into plain language so organizations can make informed decisions.

Their responsibilities include:

- Identifying correlations.
- Finding patterns.
- Applying statistical methods to analyze and mine data.
- Visualizing data to interpret and present findings.

1.2.2 Data Scientist

Data scientists:

- Analyze data to generate actionable insights.
- Build Machine Learning (ML) and Deep Learning (DL) models.
- Train models on historical data to create predictive models.

1.2.3 Data Engineer

Data engineers:

- Extract and organize data from different sources.
- Clean and prepare data.
- Store and manage data in repositories so other data professionals can access it efficiently.

1.2.4 Other Data Professionals

Other professionals in the data ecosystem include:

- Business Analysts
- Business Intelligence (BI) Analysts

These professionals make business decisions based on data analysis results.

Regardless of their specific roles or job titles, data science professionals generally work collaboratively to extract and transform data needed to make forecasts and predictions using Machine Learning (ML) and Artificial Intelligence (AI).

1.3 What Does a Data Scientist Do?

Data scientists perform a variety of responsibilities, including:

- Working with large amounts of structured and unstructured data.
- Cleaning and transforming ingested data so it becomes suitable for Machine Learning, Deep Learning, and AI models.
- Building and applying Machine Learning, Deep Learning, and Artificial Intelligence (AI) models to solve business problems.

By combining knowledge of statistics, business logic, and computer programming, data scientists design, create, and implement models that use disparate data sources (data collected from different and unrelated sources).

These models help organizations with:

- Corporate goal planning.
- Business decision-making.
- Predictions.
- Forecasting using information synthesized from multiple data sources.

1.4 Examples of Business Questions Solved Using ML and AI

Machine Learning and AI can help answer questions such as:

- How many new social media followers am I likely to get next month?
- What percentage of my customers am I likely to lose to the competition in the next quarter?
- Is this financial transaction unusual for this customer?

1.5 Skills Required for a Data Scientist

A successful data scientist requires a combination of technical knowledge and domain expertise.

1.5.1 Technical Skills

Data scientists need:

- A solid foundation in statistics and probability.
- Knowledge of Machine Learning.
- Cloud computing skills.
- Programming skills.
- Database knowledge.
- Data modeling skills.

Higher mathematics such as:

- Calculus
- Linear Algebra

are not strictly necessary but can be useful assets.

1.5.2 Domain Knowledge

Domain-specific knowledge is also valuable.

For example, if we are interested in the medical field, knowledge of:

- Research methods
- Biology

can make us stronger candidates.

1.6 Programming Languages and Tools

1.7 Python

Python is a vital tool for data scientists.

A survey published by **Kaggle** showed that:

- **79.9%** of data science professionals advise beginners to learn Python first.
- **86.7%** reported that Python is the programming tool used most frequently in their workplace.

1.8 Other Programming Languages

Someone entering the data science field may also benefit from learning:

- SQL
- R
- C++
- Java

1.9 Big Data Technologies

Skills in the following technologies are also in high demand:

- Hadoop
- Apache Spark

1.10 Business Intelligence and Visualization Tools

Useful tools include:

- Power BI (Dashboard design and creation)
- Tableau (Data visualization and storytelling)

1.11 Essential Soft Skills

In addition to technical expertise, data scientists require several important soft skills.

These include:

- Communication and presentation skills to effectively convey insights derived from datasets.
- Critical-thinking skills to analyze and solve problems.
- Creative thinking skills to develop innovative approaches and useful models.
- The ability to convert data into a story that organizations can understand and apply to their goals.
- A collaborative approach because data scientists regularly work with other data professionals.
- Diligence and tenacity to complete projects successfully, especially when they become complex or unexpected challenges arise.

1.12 Starting a Career in Data Science

If we are new to the field of data science, our first position in an organization is likely to be an entry-level role.

However, experience from other fields should also be considered an important asset.

Many skills are transferable, especially those gained through previous technical work.

Even hobbies and personal interests may provide valuable experience and expertise that can improve marketability, depending on the company or industry.

Therefore, it is important to evaluate our background and identify capabilities that can be applied in a data science career.

1.13 Career Growth Opportunities

As we gain experience, we can advance our careers in several different directions.

Depending on our interests and goals, we may choose to:

- Become a Machine Learning specialist.
- Advance our programming expertise.
- Develop deeper data analysis skills.
- Become a manager of a data science team.
- Work as an independent consultant.
- Pursue many other specialized career paths.

1.14 Key Takeaways

- Data scientists apply specialized skills and expertise to discover patterns and make predictions from large datasets.
- A career in data science requires specialized knowledge along with strong technical skills.
- Data scientists have multiple career growth opportunities as they gain experience.
- Data science is one of the fastest-growing professions in the world today.

2 Build Your Portfolio

2.1 Learning Objectives

After studying this topic, we will be able to:

- Identify the importance of a portfolio.
- Explain how to build a portfolio.
- List projects to add to a portfolio.

2.2 Importance of a Portfolio

Studying can help us in getting a job, but we will have to put in some work as well.

Many courses will equip us with job-oriented skills, and we can certainly use their certificates to enrich our resume. We can also display our knowledge during an interview.

However, a professional portfolio is an effective tool to highlight our skills and demonstrate our ability to grow, and an employer can take time to examine it carefully.

Employers are looking for a range of skills, including:

- Technical abilities
- Communication skills
- An understanding of good business practices

All of these can be demonstrated in a portfolio.

2.3 Projects to Include in a Portfolio

Examples of relevant previous work include:

- Research work or a capstone project completed as part of a degree program.
- Freelance work, provided we have permission to include it.
- Internship experience and projects, provided we have permission to showcase them.
- Projects developed while acquiring additional data science skills.
- Projects that have been launched in a workplace or internship. If possible, mention our contribution and include a link to the project.
- Projects built around the key skills mentioned in the job descriptions we are targeting.

2.4 Additional Ideas for Portfolio Content

When we want to understand something on a deeper level, building on the success of predecessors is always a great idea.

For example:

- [Coding a neural network from scratch using a research paper.](#)

This demonstrates:

- [The ability to understand research papers.](#)
- [The ability to implement our own algorithm without relying on a predefined library.](#)

Another excellent way to strengthen a portfolio is by participating in data science competitions hosted on platforms such as:

- [Kaggle](#)
- [HackerRank](#)
- [LeetCode](#)

These competitions are normally focused on real-world problems. Participating in them and documenting our work in the portfolio can be promising, especially if we rank highly in the competition.

If we create content in the form of a blog or vlog, we should include links to it. This is a terrific way to demonstrate:

- Soft skills
- Technical knowledge
- The ability to share knowledge with other professionals

Course projects and hobby projects can also become valuable portfolio projects by adding more details and features, such as:

- Error logging
- Error handling

- Additional functionality that improves the project

2.5 Steps to Build a Strong Portfolio

2.5.1 Understand the Required Skills

To build a portfolio for the job we want, we should first understand which skills we need to demonstrate.

Studying current job listings typically shows which skills are best highlighted in portfolio projects.

2.5.2 Generate Project Ideas

When generating our own project ideas, we should think about:

- Issues we are curious about.
- How relevant datasets could help shed light on a problem.

We can also find inspiration by visiting data science communities and exploring projects created by other people.

2.5.3 Collect a Real-World Dataset

Once we decide on a project idea, we need data.

It is recommended to use a real-world dataset because real datasets are often messy and require data cleaning.

Useful resources for public datasets include:

- [Data.gov](#)
- [Reddit Datasets](#)
- [GitHub's Awesome Data](#)
- [Google Dataset Search](#)

2.5.4 Clean the Data and Perform Exploratory Data Analysis (EDA)

After selecting a research question and dataset, we should:

- Clean the data.
- Perform Exploratory Data Analysis (EDA).

This stage provides opportunities to demonstrate:

- Creative thinking
- Problem-solving skills
- Exploration from multiple perspectives
- Finding supplemental data sources
- Creating compelling visualizations

After obtaining clean data, we can begin applying machine learning models to solve the problem.

2.5.5 Host Projects on GitHub

It is smart to maintain several projects that demonstrate the breadth of our skills and interests on GitHub.

GitHub is accessible and familiar to most professionals in technical fields.

We can:

- Deploy our code using GitHub Pages.
- Use basic HTML templates to make projects look professional.
- Include a well-written **README.md** file describing the project and explaining the results to demonstrate effective communication skills.

2.6 Key Takeaways

- A portfolio is an excellent way to showcase our work and skills because employers value what they can directly see and evaluate.
- Employers can better understand our practical abilities through our portfolio.
- Many different types of work can be included to strengthen a portfolio.
- Building new and increasingly complex projects helps make a portfolio stronger.
- A strong resume is important for job seekers, but a portfolio provides practical, hands-on evidence of our skills and abilities.

3 Drafting an Effective Resume

3.1 Learning Objectives

After completing this topic, we will be able to:

- Compose a basic resume.
- Determine what should be included in a resume.
- List skills and experience appropriately with good organization.
- Identify how to optimize a resume for Applicant Tracking Systems (ATS) and search engines (SEO).

3.2 Introduction

Technical fields offer many job opportunities, but they are also highly competitive. To secure the role we want, we need to market ourselves effectively and develop a plan that helps us stand out from other applicants.

Regardless of our job search approach, we need a well-written resume that presents our skills and experience concisely and effectively.

Our resume, together with our portfolio, creates the first impression that potential employers have of us. Just like a portfolio, a resume should be carefully designed to showcase our strengths and competence in the field.

Contrary to what many people believe, a resume should **not** be a paper version of a portfolio. Instead, the portfolio and resume should complement each other. **Our resume should also be closely aligned with our LinkedIn profile, and all three should contain consistent information.**

3.3 Resume Writing Best Practices

Before writing a resume, keep the following recommendations in mind:

- Keep the resume brief.
- If we have extensive relevant experience, two pages are acceptable, but avoid exceeding two pages.
- When using bullet points, make them informative yet concise.
- Use the **present tense** for our current position.
- Use the **past tense** for previous positions and completed projects.
- Carefully check spelling and grammar.
- Have someone proofread the resume if possible.
- Double-check all contact information.
- Use a professional-looking email address.

3.4 Resume Sections

3.4.1 Work Experience

The **Work Experience** section is usually the largest part of a resume.

- It may appear before or after the **Projects** section, depending on which area we want to emphasize.
- Recent graduates may choose to highlight the **Education** section if it contains substantial relevant information.
- The order of sections may also vary depending on the resume template.

When describing work experience, follow this structure:

- Begin with an **action word**.
- Describe the **task or responsibility**.
- End with the **result**, preferably using measurable outcomes.

Example:

Deployed improved database design, which increased efficiency by **25%** and saved **15%** in storage.

Some effective action words include:

- Initiated
- Constructed

- Converted
- Deployed
- Led
- Designed

We may also use other appropriate action verbs.

3.4.2 Projects

If we are new to the job market, including a **Projects** section is highly beneficial.

This section allows us to showcase projects that may not fit into a formal job role.

As we gain more professional experience, this section may become unnecessary.

3.4.3 Skills

A dedicated **Skills** section highlights our technical abilities that may not be obvious from previous job descriptions.

When creating this section:

- Include only technical skills relevant to the position we are seeking.
- List programming languages, software, tools, frameworks, and technologies.
- Initially include all relevant skills.
- Before applying for a specific role, review the job description and remove irrelevant skills.
- Add a few additional skills that differentiate us from other candidates.
- [Avoid overwhelming employers with an excessively long skills list.](#)

Depending on the resume template, skills may be presented:

- As a simple list.
- As graphical indicators showing proficiency levels.

3.4.4 Education

Include educational qualifications using the following format:

- Degree Type and Major
- University Name
- Year Graduated
- GPA and any honors (optional)

3.4.5 Hobbies and Interests

Relevant hobbies and interests can strengthen a resume under certain circumstances.

They should generally be included only when:

- We have limited work experience.
- We have limited educational experience.
- We possess limited job-related skills.
- The job posting includes responsibilities that align with our hobbies or interests.

If we already have significant professional experience and technical expertise, hobbies and interests are generally unnecessary and may even become distracting or overly personal.

3.4.6 Additional Sections

Additional sections may include:

- Conferences
- Publications
- Awards
- Technical Training
- Certifications

Include these sections only when they are relevant to the position. Avoid adding them merely to increase the length of the resume.

3.5 Communication Skills

Excellent communication and presentation skills complement the technical skills required for success in data science.

As modern publishing software and professionally designed templates become more accessible, creating an attractive resume has become much easier.

However, visual appeal alone is not sufficient.

3.6 Creating an ATS-Friendly Resume

In addition to relevant content and visual appeal, a resume should be easy for **Applicant Tracking System (ATS)** software to read.

ATS software is widely used in technical fields to automate the process of screening thousands of job applications.

The software searches resumes for relevant keywords and advances qualified candidates further in the hiring process.

Unfortunately, resumes containing:

- Multiple columns
- Images
- Creative layouts
- Complex designs

are often not read correctly by ATS software, which may prevent qualified applicants from being considered.

3.6.1 ATS-Friendly Resume Tips

- Match terminology with the language used in job postings.
- Use common industry terms.
- Avoid inserting keywords randomly, as ATS software can detect keywords used out of context.
- Keep the resume easy to scan.
- Organize information logically.
- Keep descriptions concise.
- Place contact information at the top of the page.

3.7 Optimizing a Resume for Search Engines (SEO)

Search Engine Optimization (SEO) is the process of writing web content so that it ranks higher in search engine results.

We can improve the online visibility of our resume by incorporating both SEO and ATS-friendly practices.

In many cases, the techniques that improve ATS readability also improve search engine discoverability.

3.7.1 SEO Best Practices

- Use keywords naturally throughout the resume.
- Do not simply place keywords together in a list.
- Include the phrases and terminology that search algorithms are designed to detect.
- Stay current with emerging technologies, terminology, and commonly used acronyms.
- Refer to technologies using standard industry terminology.
- Include relevant links and verify that they work correctly.
- Consider adding links to professional social media profiles if they support our professional brand.
- Ensure all public social media content is appropriate for prospective employers.

3.8 Accessibility Best Practices

To make our resume and portfolio more accessible:

- Avoid using large image files.
- Add **alt-text** (alternative text that describes images for screen readers and visually impaired users) to every image, photo, and graph.
- Alt-text also contributes additional searchable keywords.

3.9 Resume File Formats

Most ATS software performs best with **Microsoft Word**, and many employers require resumes in this format.

Therefore, we should always maintain a Microsoft Word version of our resume.

Additional resume formats may include:

- Graphic
- Plain Text
- Braille
- PDF
- Accessible PDF
- Versions in languages other than English

3.10 Key Takeaways

- A resume creates the first impression for potential employers, making its quality extremely important.
- Effective resumes follow established best practices for organizing and presenting information.
- ATS and SEO systems are widely used, so resumes must be optimized for both scanning and searchability.
- A resume is our primary tool for obtaining job opportunities.
- Investing time in creating a strong foundational resume allows us to tailor it efficiently for different job applications.

4 Expert Viewpoints: Attention-Getting Resumes

4.1 Learning Objectives

After studying this topic, we will be able to:

- Understand the features of an attention-getting resume and cover letter.
- Learn how to tailor resumes and cover letters for specific job roles.
- Identify the qualities hiring managers look for in data science candidates.
- Understand the importance of practical experience and professional portfolios.
- Recognize the skills and experiences required for a successful data science career.

4.2 What Hiring Managers Look for in a Resume and Cover Letter

Hiring managers evaluate resumes and cover letters differently.

- The **resume** should clearly present our background, experience, and qualifications.
- The **cover letter** should explain **why we are the right candidate for the specific role**.

The cover letter should be personalized for each application by discussing the skills and attributes requested in the job description. A generic cover letter is a major disadvantage because it does not demonstrate genuine interest in the position.

4.3 Common Resume Mistakes

Hiring managers frequently encounter avoidable mistakes, including:

- Spelling mistakes.
- Typographical errors.
- Grammatical errors.
- Missing basic information, such as the applicant's name.

Before submitting a resume, we should carefully review every detail to ensure it is accurate and professional.

4.4 Resume Design and Presentation

There are many resume templates available through platforms such as:

- LinkedIn
- Microsoft Word

When selecting a resume design, we should tailor it to the role we are applying for.

- If the role values creativity, we can demonstrate some creativity and personality.
- If the role is more formal, we should avoid overly flashy or distracting designs.

A resume creates the first impression, so it should clearly communicate:

- Who we are.
- What we do.
- Why we are the right candidate.

If these three points are not communicated effectively, the resume has not achieved its purpose.

4.5 Aligning the Resume, Cover Letter, and LinkedIn Profile

Our resume should provide an accurate profile of who we are and the work we have completed.

It is important that:

- Our resume and LinkedIn profile are consistent.
- The resume highlights the important requirements mentioned in the job description.
- The cover letter addresses the specific requirements of the advertised role.
- The resume, cover letter, and LinkedIn profile all align with each other.

This alignment helps hiring managers understand why we are a good fit based on our experience and training.

4.6 Building Practical Experience

Practical experience is often more valuable than simply listing years of experience.

If we are just starting our careers, we should gain experience through activities such as:

- Participating in hackathons.
- Completing internships.
- Entering Kaggle competitions.
- Building our own projects.
- Publishing project articles on a blog platform.
- Publishing source code on GitHub.
- Recording YouTube videos to demonstrate our work.

These activities help showcase practical skills and commitment to continuous learning.

4.7 Writing an Effective Cover Letter

A strong cover letter should directly address the job description.

For each requirement listed in the job description, we should:

- Explain where we gained the required experience.
- Describe how our previous work prepared us for the role.
- If we do not yet possess a particular skill or experience, explain how we plan to acquire it.

This demonstrates initiative and a willingness to learn.

4.8 Keeping the Resume Concise

A clean, one-page resume is generally considered the gold standard.

Many professional templates are available online, allowing us to choose one that matches our style.

Resumes longer than one page may not be read completely because they can indicate that the applicant is unable to communicate information concisely.

4.9 Tailoring the Resume for Different Roles

Relevant experience should match the position being applied for.

For example:

- If applying for a **Machine Learning** position, the resume should emphasize machine learning projects and experience.

- If the resume mainly highlights data visualization projects, it may not present us as the strongest candidate for a machine learning role.

For this reason, it is often beneficial to create multiple versions of a resume for different career paths.

Examples include:

- **Business Data Science Resume**
 - Communication skills.
 - Data visualization.
 - Business delivery and stakeholder interaction.
- **Machine Learning or Statistics-Focused Resume**
 - Machine learning expertise.
 - Statistical knowledge.
 - Technical skills and model development.

Each resume should emphasize the skills that best match the target position.

4.10 Important Skills and Experiences for Data Science Careers

Strong data science candidates should develop expertise in several areas.

4.10.1 Technical Skills

- Mathematics.
- Statistics.
- Python programming.
- SQL.

Professional certifications, such as IBM certification programs, are also considered valuable.

4.10.2 Practical Experience

Students should actively seek opportunities such as:

- Internships.
- College-supported placements.
- Research opportunities.
- Teaching assistant opportunities.
- Voluntary learning activities.
- Data science meetups.
- Professional networking sessions.

These experiences help develop practical knowledge and professional connections.

4.11 Recommended Learning Path

A strong learning path for aspiring data scientists includes:

- Build a theoretical foundation through university courses or Massive Open Online Courses (MOOCs), such as:
 - Coursera
 - edX
- Replicate published research papers by recreating the methods and code described in the papers.
- Participate in Kaggle competitions.
- Gain practical experience by solving real-world problems.
- Collaborate with colleagues on projects.
- Participate in hackathons.
- Apply for junior-level data science positions.
- If a junior position is unavailable, begin with trainee or internship positions to build professional experience.

4.12 Key Takeaways

- Personalize every cover letter for the specific role.
- Ensure the resume, cover letter, and LinkedIn profile are consistent.
- Tailor the resume according to the job description.
- Highlight relevant experience for each position.
- Keep the resume concise, preferably one page.
- Develop practical experience through internships, hackathons, Kaggle competitions, personal projects, GitHub, blogs, and YouTube.
- Build strong foundations in mathematics, statistics, Python, and SQL.
- Strengthen theoretical knowledge through university courses or MOOCs such as Coursera and edX.
- Continue building practical experience to become competitive for junior or internship data science roles.

5 Draft Your Resume

5.1 Learning Objectives

After completing this practice exercise, we will be able to:

- Create a basic resume.
- Describe how to edit the resume to fit different job applications.

5.2 Prerequisites

We will need the following to complete this practice exercise:

- Word processing software.
- A source of information about our past work experience and skills.
- An example job listing to practice tailoring the resume to a specific role.

5.3 Important Considerations

- Keep a copy of the resume in **Microsoft Word (.doc)** format because it is the format most companies request.
- Save another copy in **plain text** format to make it easier to copy and paste information into online job application forms.
- Keep the resume **brief and concise**. It should very rarely exceed **two pages**.
- Use an example job listing to practice tailoring the resume for different positions.
- Highlight the sections that are likely to change for different job applications so they can be updated quickly in the future.

5.4 Should We Use a Resume Template?

Many resume templates are available online and can simplify resume creation while providing an attractive layout. However, use caution because many templates are **not compatible with Applicant Tracking Systems (ATS)** (software used by employers to automatically scan and filter resumes).

If a potential employer uses ATS, the software may fail to properly read a highly formatted template, reducing the chances of the resume being considered. It is better to create a resume in a simple format first. We can always copy the content into a visually appealing template later if needed.

5.5 Contact Information

Begin by creating a new document in a word processor.

Include the following information at the top of the resume:

- Write the full legal name, or a professional nickname if that is commonly used in professional settings.
- Place the name at the top of the page. It is usually centered, although left alignment is also acceptable.
- Add contact information directly below the name, including:
 - Phone number
 - Professional email address
- A street address is optional, but including the town and state can be helpful.
- Carefully verify that the phone number and email address are accurate.
- Ensure the email address appears professional.
- Include links to professional social media profiles or websites if they support professional networking.
- Review all online profiles to ensure they contain information appropriate for potential employers.

5.6 Resume Summary

The top section of the first page is the most important part of a resume because hiring managers often spend only a few seconds reviewing each application before deciding whether to continue reading.

Begin with a one-sentence professional summary describing who we are and what type of position we are seeking.

Example:

Data science graduate looking for an entry-level position in the field of medicine and health care.

Include a **Technical Skills** section listing the relevant skills we possess.

This section should be customized for each job application and may include:

- Machine learning expertise
- Programming languages
- Software tools
- Frameworks
- Databases
- Other technical or professional skills relevant to the position

5.7 Professional Experience

The Professional Experience section is typically the largest part of the resume.

5.7.1 Organizing Work Experience

- Start with the current or most recent position.
- Include:
 - Job title
 - Company name
 - Employment start date
 - Employment end date

5.7.2 Writing Experience Statements

Use the following structure for each accomplishment:

- **Action word**
- **Task performed**
- **Quantifiable result (whenever possible)**

Some useful action words include:

- Initiated

- Constructed
- Converted
- Deployed
- Led
- Designed

Example:

Deployed improved database design, which increased efficiency by **25%** and saved **15%** in storage.

5.7.3 Additional Guidelines

- Include the major responsibilities and duties performed in each position.
- Focus on the most relevant and impactful accomplishments.
- Keep the resume concise and under **two pages**.
- List previous positions in **reverse chronological order**.
- If we have **more than 8–10 years** of experience, include only the most recent and relevant positions.

5.8 Education

Include all relevant educational credentials.

For each educational qualification, provide:

- Educational institution
- Degree earned
- Major or specialization

We may also include:

- Significant areas of study
- Relevant coursework

Dates are optional. While they can provide context, they may also reveal age and potentially contribute to discrimination, so omitting them is acceptable.

Also include additional educational achievements such as:

- Professional certifications
- Standalone courses
- Other relevant educational accomplishments

5.9 Optional Resume Sections

If we have limited professional experience, additional sections can strengthen the resume.

5.9.1 Projects

Include independent or personal projects that demonstrate practical skills.

Whenever possible, include links to:

- [Portfolio](#)
- [GitHub repositories](#)
- [Project websites](#)
- [Demonstrations](#)

5.9.2 Additional Professional Activities

If relevant, include:

- Conferences attended
- Professional publications
- Awards and recognitions

5.9.3 Hobbies, Interests, and Volunteer Work

Include these sections only when they are directly relevant to the position being applied for.

5.10 Resume Tips

When preparing a professional resume:

- Do **not** include a photograph because it may unintentionally contribute to discrimination.
- Do **not** include references on the resume. References may change depending on the position and should be provided separately when requested.
- Ask a trusted friend or colleague to proofread the resume before submitting it. Employers may reject resumes containing spelling or grammatical mistakes.
- Save the resume in cloud storage services such as **Google Docs** or **Dropbox** so it can be accessed easily from any device.

6 Create Your Cover Letter

6.1 Learning Objectives

After completing this practice exercise, we will be able to:

- Create a basic cover letter.
- Describe how to edit the letter to fit different job applications.

6.2 Introduction

One of the most important tasks when beginning a job search is to develop a basic cover letter. It is a good idea to draft a basic cover letter before applying for jobs. Although we will customize the letter for each position and application, having a well-written template makes the application process much easier.

A cover letter works alongside a resume to introduce us to a prospective employer. It should briefly introduce who we are and explain why we are a strong candidate for a particular position. Instead of repeating all of our qualifications, we should highlight only the most relevant information. Detailed qualifications should remain in our resume and LinkedIn profile. The objective of the cover letter is to generate enough interest for the employer to invite us for an interview.

6.3 Prerequisites

We will need the following to complete this practice exercise:

- Word processing software.
- A source of information about our past work experience (a resume might be useful).
- An example job listing to practice tailoring the letter to a specific role.

6.4 Keep in Mind

- We will probably want to keep a copy of our cover letter in **Microsoft Word (.doc)** format because it is the format companies request more often than others.
- It is also useful to save the letter in **plain text** format because it makes it easier to copy and paste into website application fields.
- We can complete this practice exercise using pen and paper, but keeping the letter in electronic form makes it much easier to edit and reuse as a template.
- We can use our example job listing to fill in information such as the job title, qualifications, contact person, and other details.
- We should edit the letter whenever we tailor it to a different job listing.
- As we create the letter, we should highlight the sections that are likely to change, such as the job title, company name, or contact person. This makes future updates much easier.

6.5 Begin the Cover Letter

Open a new document in our preferred word processor.

Choose an opening for the letter and type it in.

Example:

Dear (Manager):

Note: (Manager) is a placeholder. When preparing an actual cover letter, replace it with the name of the hiring manager or contact person for the specific job application.

6.6 Write the First Paragraph

The first paragraph should:

- State the title of the position we are applying for.
- Explain why we believe we are a strong fit for the role.
- Mention where we found the job posting or who referred us, if applicable.
- Express enthusiasm for the company whenever possible.

Example:

I was excited when my colleague Janna Navarro told me your company had an opening for a Data Analyst. I have admired the products developed by Torotoni Software for many years, and I would be very interested in joining your team.

I am a data analyst with 3 years of professional experience and skills that match your needs, and I think I would be a strong candidate.

The final sentence serves as a transition into the next paragraph, where we expand on our qualifications.

6.7 Write the Main Body of the Letter

The main body should build a strong case for why we are a suitable candidate.

We should:

- Highlight examples from our experience that match the skills listed in the job description.
- Draw connections between the job listing, our resume, and our portfolio.
- Quantify our accomplishments whenever possible.
- Keep the content concise and focused.
- Save additional examples for discussion during the interview.

Example:

In my most recent role, I maintained and extended a mission-critical database and prepared a weekly report of changes and trends for stakeholders. I also worked with the coding and management teams to improve functionality of the database by 45%, advised on compliance issues, and supervised the work of two interns. I know SQL and Python, have intermediate proficiency in Hadoop and Apache Spark, and have used Tableau software for visualization.

6.8 Close the Cover Letter

The closing section should remain simple and professional.

We should:

- Restate our interest in the position.
- Express our desire to discuss the opportunity further.
- Include a call to action.

- End with a polite closing followed by our full name.

Example:

Thank you for your consideration. I would love to bring my skills to Torotoni Software, and I'm eager to discuss the contributions I can make to your team. Please call me at your earliest convenience.

Best regards,

Olivia Torvald

6.9 Tips

- Ask a friend or colleague to proofread the letter before sending it. A polished cover letter creates a stronger first impression, and some employers may remove applicants from consideration if the letter contains numerous mistakes.
- We can reuse the cover letter for different application methods, including:
 - Email.
 - Printed business letters.
 - Online job application portals.
- If sending the cover letter by post, format it as a professional business letter by including:
 - Our address.
 - The employer's business address.
 - Proper business letter formatting (consult an appropriate reference if needed).
- Save the cover letter in a cloud storage service, such as **Google Docs** or **Dropbox**, so it can be easily accessed from any device.

7 Expert Viewpoints: Standing Out from the Crowd

7.1 Learning Objectives

After completing this topic, we will be able to:

- Identify the technical skills that distinguish strong data science candidates.
- Understand the importance of mathematical and statistical foundations in data science.
- Recognize the role of programming, databases, and cloud technologies in data science careers.
- Explain why soft skills are equally important alongside technical expertise.
- Understand the value of continuous learning, research, and perseverance in a data science career.

7.2 What Sets Apart a Great Data Science Candidate?

Data science professionals emphasize that standing out from other candidates requires a combination of **technical expertise**, **analytical thinking**, **business understanding**, and **strong interpersonal skills**.

7.3 Technical Foundations

A strong data science candidate should possess a solid understanding of the mathematical concepts behind machine learning and deep learning.

- We should have a relatively deep understanding of the mathematics used in machine learning and deep learning.
- We do **not** have to become experts in linear algebra or solving differential equations.
- However, we should understand how machine learning algorithms work internally.

A highly recommended learning path includes:

- Machine Learning courses on Coursera.
- Deep Learning AI specialization.
- The Introduction to Deep Learning specialization consisting of **five to seven courses**.

One major advantage of these courses is that we are required to implement machine learning algorithms using **NumPy**, providing a strong practical foundation.

Although this level of mathematical understanding is **not required for every data science role**, it provides an excellent foundation for long-term growth.

7.4 Data Acquisition and Data Transformation Skills

Data acquisition and transformation skills are equally important.

Essential knowledge includes:

- Fundamental SQL knowledge.
- Understanding of databases.
- Shell scripting.
- Big data tools.

Recommended technologies include:

- Apache Spark
- Apache Spark SQL
- Spark ML (spark.ml)

However, for many use cases, the following tools are sufficient:

- Scikit-learn
- R

For deep learning, commonly used frameworks include:

- Python
- TensorFlow
- Keras

These technologies continue to gain widespread adoption across the industry.

7.5 Technical Skills and Non-Technical Skills

Success in data science is not determined by technical knowledge alone.

Employers also evaluate:

- How well we answer technical and behavioral questions.
- How methodical our approach is.
- What it would be like to work with us on a team.

When projects become challenging, team members should be able to rely on us and trust us.

Building rapport with coworkers is therefore an important professional skill.

7.6 Research, Mathematics, Statistics, and Programming

A successful data science career is built on a combination of:

- Research methodology.
- Mathematics.
- Statistics.
- Programming.

Helpful areas of study include:

- Research methodology involving mathematics and statistics.
- SQL.
- Python programming.

Experience with data analysis and visualization tools also strengthens our profile.

Examples include:

- Tableau
- MicroStrategy

Overall, an ideal combination includes:

- Data analysis experience.
- Data visualization tools.
- Mathematical methods.
- Statistical methods.
- Python programming.
- SQL.

7.7 Database, Cloud, and Analytics Skills

A great data science candidate should have strong knowledge of databases and programming languages.

Important technical skills include:

- Database concepts.
- SQL.
- Python for data science.

Python is especially valuable because it can be used across many different areas of data science.

Cloud computing knowledge is becoming increasingly important.

We should understand:

- Data lakes (centralized repositories that store structured, semi-structured, and unstructured data at scale).
- Data warehouses (systems designed for storing and analyzing structured business data).
- Managing data in cloud environments.

These cloud-related skills are becoming increasingly critical in modern data science roles.

7.8 Statistics and Business Intelligence Tools

Statistics plays a major role in advanced analytics, including:

- Predictive analytics.
- Diagnostic analytics.
- Prescriptive analytics.

Common business intelligence and analytics tools include:

- Microsoft Excel
- Tableau
- Looker
- Power BI

These tools represent many of the hard skills required to succeed in data science roles.

7.9 Essential Soft Skills

Technical skills alone are not enough.

Successful data scientists also develop strong soft skills.

Important soft skills include:

- Business analysis.
- Stakeholder communication.
- Strategic communication.
- Research skills.
- Grit and perseverance.

7.9.1 Business Analysis

Business analysis involves understanding problems from a business or domain perspective.

For example:

- The approach used in the financial sector may differ from the approach used in the healthcare sector because each domain has different requirements and objectives.

Understanding the business context helps us develop more effective solutions.

7.9.2 Stakeholder Communication

We should be able to:

- Communicate technical ideas clearly.
- Explain our analytical approach.
- Engage effectively with stakeholders.
- Deliver solutions strategically.

Strong communication is critical to the successful delivery of data science projects.

7.9.3 Research and Problem-Solving

An effective data scientist knows how to:

- Research problems.
- Search for information efficiently.
- Learn independently.

The ability to investigate unfamiliar problems is an essential professional skill.

7.9.4 Grit and Perseverance

Grit is one of the most valuable qualities for long-term success.

Throughout a data science career, we will encounter:

- Coding challenges.
- Project setbacks.
- Difficult debugging sessions.
- Unexpected obstacles.

Ups and downs are a normal part of every data science career.

Success depends on our ability to remain persistent, continue learning, and push ourselves toward the final goal.

7.10 Key Takeaways

- A strong mathematical foundation helps us understand how machine learning and deep learning algorithms work.
- SQL, Python, databases, shell scripting, and big data technologies are essential technical skills.
- Apache Spark, Spark SQL, TensorFlow, Keras, Scikit-learn, and R are valuable technologies for data science.
- Research methodology, mathematics, statistics, and programming together create a strong foundation for success.
- Cloud technologies, data lakes, and data warehouses are becoming increasingly important.
- Business intelligence tools such as Excel, Tableau, Looker, Power BI, and MicroStrategy strengthen data analysis capabilities.
- Great data scientists combine technical expertise with communication, business understanding, research skills, and perseverance.

8 Gathering References

When preparing to look for work, it is useful to have additional materials ready alongside our resume and portfolio. Having these resources prepared in advance makes the job application process faster and easier.

In addition to a resume and portfolio, we should prepare the following materials:

- A **basic cover letter** that can be tailored for each position we apply for.
- An **elevator pitch** (a short description of who we are and what we are looking for) that can be used in various networking and interview situations.
- A **list of references** who can vouch for our past experience and qualifications.

Preparing these materials at the beginning of the job-seeking process helps us respond quickly to employment opportunities.

8.1 References

Many employers ask candidates to provide references during the interview process. Therefore, it is beneficial to collect references in advance.

Preparing references early allows us to:

- Contact our chosen references to confirm that they are willing to speak with prospective employers.
- Gather their current contact information.
- Obtain their permission to share their contact details with employers.

For each reference, we should collect the following information:

- Name
- Title
- Company
- Mailing Address
- Phone Number
- Email Address

We should keep a directory of our references and their contact information in a place where it can be easily accessed whenever needed.

8.2 Professional References

Professional references are individuals who can comment on our previous work experience and professional performance.

Typically, professional references should be managers or supervisors who have worked directly with us and can discuss our responsibilities and performance at previous jobs.

However, some managers may be restricted by company policies from discussing the details of our work. In such cases, they may only be able to verify:

- That we worked for the company.
- Our employment start date.
- Our employment end date.

Although employers generally prefer supervisors as professional references, we can also include:

- Colleagues who worked closely with us.
- Professors who are familiar with our coursework, especially if we have limited work experience.
- Individuals who supervised or worked with us on volunteer projects.

8.3 Personal References

Personal references are individuals who can speak about our personal qualities rather than our professional experience.

They may comment on characteristics such as:

- Personal character
- Trustworthiness
- Honesty
- Work ethic
- Reliability

Personal references are less commonly requested in modern interviews, but it is still worthwhile to prepare them.

Personal references should:

- Be people who know us well.
- Not be members of our family.
- Be willing to be contacted by prospective employers.

As with professional references, we should collect their current contact information and obtain their permission before sharing their details with employers.

9 Drafting Other Materials – Elevator Pitch

9.1 Introduction

An **elevator pitch** is one of the most effective tools we can use during a job search. It is a short, easily memorized statement that describes who we are, some of our key strengths as a job seeker, and our career goals. We can use it in both spoken and written situations to introduce ourselves effectively.

Although employers are unlikely to ask directly for an elevator pitch, having one prepared can help in many situations. It can assist us in:

- Drafting a cover letter.
- Introducing ourselves to potential employers while networking.
- Answering interview questions such as **"Tell us about yourself."**
- Presenting ourselves confidently and professionally during interviews.

A well-prepared elevator pitch helps us appear poised, organized, and prepared.

9.2 Key Questions an Elevator Pitch Should Answer

An effective elevator pitch should answer the following three questions:

- **Who are we?**
- **What do we do?**
- **What do we want?**

9.3 Example Elevator Pitch

"Hi, I'm Jonathan Bittmann. I'm a data scientist working toward a move into management. I've spent the last six years as a data analyst for a cloud development company, where I've developed my skills in programming and served as team leader. While leading several successful project teams, I became interested in data science management and have gone back to school to earn a business administration degree. While I'm working on that, I'm looking for a position in project management so I can put my learning to use. I'm very interested in the work your company does and I think I could bring a lot of expertise and leadership to your organization. Could we meet to discuss the open project manager roles you currently have?"

9.4 Components of an Elevator Pitch

9.4.1 Who Are We?

Begin by introducing ourselves with our **full name** and, if applicable, our **preferred name**.

If we are introducing ourselves in person, this is an appropriate time to offer a handshake.

Follow the introduction with a brief statement about:

- Our career title.
- Our current professional or educational situation.

9.4.2 What Do We Do?

Describe a few key skills or experiences that make us a strong candidate.

When selecting what to include:

- Highlight one or two of our strongest or most recent achievements.
- Avoid trying to summarize our entire resume.
- Arrange our experiences logically so they form a story instead of a list of bullet points.

Presenting our experience as a story helps engage the listener or reader more effectively.

9.4.3 What Do We Want?

Conclude the elevator pitch with an action request.

Depending on the situation, this may include:

- Asking for an interview.
- Explaining our interest in a particular position.
- Requesting a meeting.
- Asking whether our response answers the interviewer's question.

The purpose is to encourage the other person to continue the conversation rather than simply listen.

Because situations vary, it is helpful to prepare several possible endings that can be adapted as needed.

9.5 Tips for Writing an Effective Elevator Pitch

While composing an elevator pitch:

- Use natural, conversational language.
- Avoid technical jargon whenever possible.
- Remain professional while using wording that feels comfortable and authentic.

- Ensure the pitch is easy for a wide audience to understand so it can be used in many different situations.

9.6 Practicing the Elevator Pitch

After writing the elevator pitch:

- Memorize it until it becomes natural.
- Practice delivering it aloud.
- Speak slowly and clearly.
- Enunciate each word carefully.
- Remember that nervousness often causes us to speak too quickly, so consciously slow our pace.
- Practice with a friend who can provide feedback and help us improve.
- Consider developing and practicing elevator pitches together.

9.7 Tailoring the Elevator Pitch

Although the elevator pitch should be written to be broadly useful, we should be prepared to adapt it for different situations.

Possible adjustments include:

- Changing the action request.
- Highlighting different experiences.
- Emphasizing skills that are most relevant to a particular opportunity.

Tailoring the pitch to the specific context makes it more effective while keeping the overall structure the same.

9.8 Key Takeaway

An effective and well-rehearsed elevator pitch can be an extraordinarily valuable tool throughout the job search process.

Investing time in developing, memorizing, practicing, and tailoring our elevator pitch can provide significant benefits when networking, interviewing, writing application materials, and communicating with potential employers.

Module 2: Applying and Preparing to Interview

10 Company and Industry Research for Data Science

10.1 Learning Objectives

After completing this topic, we will be able to:

- Explain how to research a company we will be interviewing with.
- Recognize good sources for company and industry research.
- Discuss how to conduct an informational interview.

10.2 Introduction

When we schedule a job interview, we may immediately begin wondering how we can prepare for a great interview. The best place to start is by learning as much as possible about the organization we are interviewing with and the industry we want to enter.

10.3 Researching a Company

The company's own website is a rich source of information. The following sections are particularly valuable.

10.3.1 History

The **History** section may provide information about:

- The company's origins and history.
- Its mission and vision statements.
- The company's goals.
- The company culture.

10.3.2 Keywords

Read carefully through the website and notice repeating keywords.

The company's choice of words to describe what it does or what it represents can help us understand the image the company strives to project and maintain. We may want to use this language during our interview to show that we are in tune with the company.

10.3.3 Careers

Looking at other job listings can help us understand:

- Where the company plans to expand.
- The company's priorities and future direction.

10.3.4 Data

As data scientists, we should pay close attention to quantitative information, such as:

- Sales reports.
- Number of customers.
- Details about the scope of work accomplished.
- Company achievements.

This information may also provide clues about the tools the company uses to display and present information.

10.4 Other Sources for Company and Industry Research

Although the company website is the most readily available source to begin our research, many other valuable resources are available.

10.4.1 Chamber of Commerce and Better Business Bureau

Check our local Chamber of Commerce or the Better Business Bureau website for additional information about the company.

10.4.2 Social Media

Social media is a good way to find:

- Current company news.
- Recent activities.
- Information about the company culture.

10.4.3 Web Search

A web search may reveal valuable information if the company has been featured in publications or news reports.

Look for:

- Articles about the company.
- Articles about individuals associated with the company.

We may even discover information that makes us rethink becoming involved with a particular company.

10.4.4 LinkedIn

LinkedIn provides extensive company information.

If we know the name of our interviewer, reviewing their LinkedIn profile can help us learn about their professional background.

10.4.5 Glassdoor

Websites such as Glassdoor allow employees and customers to post reviews about companies.

Read several recent reviews to learn about:

- Employee experiences.
- Customer experiences.
- Interview reports.
- What to expect during the interview process.

10.5 Informational Interviews

An informational interview is an informal meeting with someone working in a field we want to pursue.

It is **not** a job interview. Its primary purpose is to research careers.

An informational interview can provide:

- Valuable career tips.
- Expert advice.
- Guidance for choosing a career path.

10.6 Finding Someone for an Informational Interview

To find someone to interview:

- Make a list of **three to five companies** we would like to learn more about, preferably within our area.
- Use LinkedIn or another career network to identify professionals working as data scientists at those companies.
- Use professional networks or social media to find other data scientists.
- Ask a teacher, professor, or friend to recommend someone they know.
- Prepare a brief introduction explaining the purpose of our request.
- Reach out by:
 - Email
 - Phone
 - LinkedIn
 - Another professional platform

10.7 Conducting an Informational Interview

During the interview:

- Allow the conversation to flow naturally.
- Be open to unexpected information and insights.
- Take notes, especially on useful tips and expert advice.
- Monitor the time throughout the meeting.
- Check halfway through our questions to ensure we stay within the agreed-upon time.
- Keep the interview to **30 minutes or less**.

10.8 After the Interview

Once the interview is complete:

- Review our notes.
- Highlight areas requiring further research.
- Send a thank-you note to the interviewee.
- Stay connected through LinkedIn or other professional platforms.
- Continue building our professional network.

10.9 Suggested Informational Interview Questions

10.9.1 Questions About the Interviewee's Background

- How did you become interested in the data scientist role?
- How did you prepare to enter the field?
- How did you begin your career?

10.9.2 Questions About the Current Role

- What is a typical workday like for you?
- What do you like most about your work?
- What is one of your favorite projects that you have worked on?
- What is the most difficult challenge you have dealt with?
- What would you like to accomplish next in your career?

10.9.3 Questions for Career Advice

- What related fields or career paths would you recommend we also look into?
- How do you stay current and learn new technology?
- What current issues and trends in the field should we be aware of?
- What are the most effective strategies for seeking a position in this field?

10.10 Key Takeaways

- Company websites are a good primary source of information about a company.
- Other sources can provide even more information about companies and interviewers.
- A well-planned and carefully conducted informational interview can be a valuable asset during our career search.
- When we research thoroughly before an important job interview, we enter with enough background knowledge to project confidence, impress our interviewers, and stand out among other candidates.

11 Networking Online and Offline

11.1 Learning Objectives

After studying this topic, we will be able to:

- Describe how networking is important to our job search.
- List different ways to network online.
- Explain how to meet other professionals for networking offline.

11.2 Importance of Networking

Networking is essential. It helps us:

- Develop and improve our skill set.
- Stay on top of the latest trends in our industry.
- Keep a pulse on the job market.
- Meet prospective mentors, partners, and clients.
- Gain access to the necessary resources that foster our career development.

11.3 Networking Through Online Job Platforms

The most common way to search for jobs in technical fields is through job portals or professional networking sites.

Some popular job portals include:

- Indeed
- LinkedIn
- Glassdoor
- Monster

These portals contain millions of job listings and make it easier to connect with employers who are looking for our skills and experience.

In addition to job listings, these websites offer valuable resources for job seekers, including:

- Articles on job search topics.
- Blog posts with helpful career tips.

11.4 Using LinkedIn Effectively

LinkedIn is the top professional social networking site with:

- **133 million users in the United States (U.S.)**
- Users from **more than 200 countries** around the world.

Additionally:

- **87% of recruiters use LinkedIn** as part of their candidate search.

To maximize our opportunities, we should:

- Keep our LinkedIn profile up to date.
- Highlight our previous experience in detail.

- Be comprehensive about our current skills and career objectives.
- Enable the **"Open to Work"** option so recruiters can easily find us.
- Complete LinkedIn skill assessments.
- Request colleague testimonials and recommendations.
- Take advantage of other professional features available on LinkedIn.

11.5 Exploring Additional Job Search Resources

Besides major job portals, we should also:

- Check local, state, and federal job websites.
- Find out whether our state employment agency hosts job sites or job fairs.
- Use these resources to discover additional employment opportunities.

Most job websites require us to create an account so we can:

- Save job postings.
- Save search queries.
- Receive email alerts when new jobs matching our interests are posted.

Many of these websites use an **Applicant Tracking System (ATS)** (software used by employers to collect, organize, and filter job applications), which:

- Simplifies the job application process.
- Makes it easier for recruiters to find qualified candidates.

11.6 Researching Companies

It is a good idea to research industry-leading companies directly.

A strong starting point is to:

- Visit the company's official website.
- Look for its **Careers** page.
- Search for positions that match our interests.
- Create a profile if available.
- Upload our resume so recruiters can find us.

Some companies may have open positions that have not yet been publicly listed.

While researching companies, we should also connect with people who work there.

Benefits of networking with employees include:

- Understanding the company's culture.
- Learning about the work environment.
- Building valuable professional relationships.

Networking is not only about exchanging information. It is also a way to create long-term relationships that provide mutual benefits. A professional contact may think of us when their team needs a new member.

11.7 Learning from Professionals

To learn more about professionals in our field, we can:

- Review their LinkedIn profiles and posts.
- Read biography pages on their companies' websites.
- Search for news articles about their work.
- Contact them and request an informational interview.

11.8 Using Social Media for Job Searching

Whether we are recent graduates or experienced professionals seeking new opportunities, social media can be a valuable job search tool.

Some effective strategies include:

- Posting about the type of job we are seeking and that we are available for work.
- Following companies that interest us.
- Following professional organizations in our field.
- Following leading professionals.
- Participating in online discussions.
- Asking thoughtful questions and engaging with the community.

11.8.1 Social Media Caution

Employers often review candidates' social media profiles during the hiring process.

Therefore, we should ensure there is nothing on our social media accounts that could create a negative impression with potential employers.

11.9 Networking Offline

Networking offline is equally important.

A good starting point is:

- Talking to our friends.
- Meeting their friends.
- Expanding our professional circle gradually.

Additional networking opportunities include:

- Attending local meetups and industry interest groups.
- **Searching for data science events on Meetup.com.**
- Joining professional organizations.

- Attending meetings and networking events.
- Taking adult education classes to strengthen our skills.
- Attending professional conferences.
- Volunteering for organizations that can benefit from our skills.

11.10 Best Practices for In-Person Networking

Whenever we attend networking events:

- Speak with people attending the same event.
- Practice introducing ourselves professionally.
- Avoid asking new acquaintances directly for job leads before building a relationship.

As we develop stronger professional relationships, networking contacts may naturally mention job opportunities before we even ask.

11.11 Key Takeaways

- Networking is a crucial way to discover job opportunities.
- Online job boards provide numerous employment opportunities and valuable career resources.
- LinkedIn is one of the most important professional networking platforms, with **133 million users in the U.S.**, users from **more than 200 countries**, and **87% of recruiters** using it during candidate searches.
- Keeping our LinkedIn profile current and making use of its professional features improves our visibility to recruiters.
- Many job websites use **Applicant Tracking Systems (ATS)** to streamline hiring and candidate discovery.
- Researching companies directly and connecting with their employees helps us better understand company culture and build valuable professional relationships.
- Social media is a powerful job search tool when used professionally.
- Offline networking through meetups, conferences, professional organizations, classes, and volunteering expands our professional network.
- Very often, getting a job is about **who we know as much as what we know**, so having someone help us get our foot in the door can provide a significant advantage.

12 Expert Viewpoints: Building Your Network

12.1 Learning Objectives

After studying this topic, we will be able to:

- Understand the importance of building a professional network.
- Identify online and offline platforms for networking.
- Learn how to create meaningful professional relationships.
- Discover how networking can improve career opportunities, particularly in data science.

12.2 Introduction

Building a professional network is an essential part of career development. A strong network helps us learn from experienced professionals, discover new opportunities, receive mentorship, and increase our visibility in the industry. Networking is not only about asking for help but also about contributing to the community by helping others.

12.3 Build Your Network Through Online Communities

One of the best ways to build a professional network is by participating in online communities related to our field of interest.

Some recommended platforms include:

- **Twitter (X):**
 - Follow industry leaders and professionals who regularly discuss topics related to our career interests.
 - Like their posts.
 - Reply to their posts.
 - Ask thoughtful questions.
 - Join conversations and discussions.
- **Stack Overflow and Similar Communities:**
 - Participate in technical discussions.
 - Ask and answer questions.
 - Learn from experienced developers and professionals.
- **Discord and Slack Communities:**
 - Join communities dedicated to specific technologies or topics.
 - Participate in discussions.
 - Share knowledge and learn from others.
- **Online Forums:**
 - Search for forums focused on our area of interest.
 - Engage with community members and contribute to discussions.

12.4 Attend Local Events and Meetups

Networking is not limited to online platforms. We should also attend local events where professionals gather.

Some valuable opportunities include:

- Local meetups
- Professional networking events
- Conferences
- Industry gatherings
- Technology communities
- Maker spaces
- Hacking labs
- Hackathons

These events allow us to:

- Meet professionals in person.
- Learn about current industry trends.
- Build long-term professional relationships.
- Expand our professional network.

12.5 Using LinkedIn Effectively

LinkedIn is one of the most valuable platforms for professional networking.

12.5.1 Create a Strong LinkedIn Profile

If we already have a LinkedIn profile, we have already started building our professional network.

To improve our profile:

- Follow LinkedIn's guide to achieve an **All-Star Profile**.
- Complete every section of the profile.
- Add our key skills.
- Include projects we have worked on.
- Request recommendations from colleagues and mentors.
- Add relevant videos and useful links when appropriate.
- Make the profile visually appealing and interesting.

12.5.2 Connect with Relevant Professionals

Building a large and relevant network can significantly help during job searching.

We should connect with:

- Recruiters
- Previous colleagues
- Managers
- Industry professionals
- Mentors
- Experts in our field

Focus on building relationships with people who are relevant to our career goals.

12.5.3 Personalize Connection Requests

When sending LinkedIn invitations:

- Avoid using the default invitation message.
- Add a few personalized words explaining why we would like to connect.
- Personalized invitations feel much more genuine and increase the likelihood of acceptance.

12.6 LinkedIn and Data Science Careers

Recruiters frequently use LinkedIn to identify qualified candidates, especially for **Data Science** positions.

An effective LinkedIn profile can:

- Increase our visibility to recruiters.
- Demonstrate our skills and experience.
- Showcase our professional achievements.
- Improve our chances of being considered for relevant positions.

LinkedIn also allows us to:

- Request professional references.
- Give recommendations to others.
- Build credibility through endorsements and recommendations.

Professional references can describe:

- Our previous work.
- Our experience.
- Our professional strengths.

These references help recruiters evaluate our profile more carefully and determine whether we are a good fit for available positions.

12.7 Attend Conferences and Professional Events

Professional conferences provide excellent opportunities to expand our network.

Examples include:

- Gartner conferences.
- Professional conferences related to our industry.
- Local professional gatherings.
- Meetups.

Attending these events helps people become familiar with us and increases our professional visibility, making us more marketable for data science careers.

12.8 Reach Out to Professionals

Networking sometimes requires taking the initiative.

One effective approach is to:

- Contact professionals on LinkedIn.
- Invite them for a coffee or an informal conversation.

- Learn from their experiences.

This approach may require persistence.

For example:

- We may need to send **10 to 20** connection requests to receive **1 or 2** positive responses.

Even a few successful conversations with experienced data scientists from large companies can provide valuable insights and career guidance.

12.9 Build Relationships Through Events

Networking is about creating long-term relationships rather than collecting contacts.

By attending different events across multiple industries, we can:

- Understand different career paths.
- Learn about various industries.
- Expand our professional perspective.
- Build relationships that may become valuable immediately or even several years later.

We never know whether someone in our network will be able to help us today or five years from now.

12.10 Share Knowledge and Help Others

Networking is a two-way process.

We should not focus only on receiving help.

Instead, we should:

- Answer questions from beginners.
- Share useful resources.
- Offer guidance whenever possible.
- Contribute to professional communities.

Helping others strengthens our reputation and encourages people to support us in return.

12.11 Additional Online Networking Platforms

Besides LinkedIn, other useful networking platforms include:

- [Twitter \(X\)](#)
- [Meetup groups](#)
- [GroupMe](#)
- [Discord](#)
- [Slack](#)

- [Stack Overflow](#)
- [Online technical forums](#)

On these platforms, we should:

- Talk about what we do.
- Share what we enjoy learning.
- Share useful resources.
- Participate in discussions.

When we consistently contribute valuable content, professionals naturally begin reaching out to us.

12.12 Key Takeaways

- Building a professional network is essential for long-term career success.
- LinkedIn is one of the most effective platforms for networking and job opportunities.
- Complete and maintain an engaging LinkedIn profile.
- Personalize LinkedIn connection requests instead of sending generic invitations.
- Connect with recruiters, colleagues, managers, mentors, and industry professionals.
- Participate in Twitter (X), Stack Overflow, Discord, Slack, GroupMe, and online forums.
- Attend conferences, meetups, hackathons, maker spaces, and professional gatherings.
- Request and provide professional recommendations on LinkedIn.
- Be proactive by reaching out to professionals for conversations or coffee meetings.
- Be persistent—sending **10–20** requests may result in **1–2** meaningful responses.
- Share knowledge, answer questions, and contribute to the community.
- Strong professional relationships can create opportunities both now and in the future.

13 Assessing Job Listings

13.1 Learning Objectives

After completing this topic, we will be able to:

- Dissect and read job listings critically.
- Describe the differences between Full-Time Employee (FTE) and contract positions.
- Navigate company application policies.
- Recognize job listings that may be problematic.

13.2 Understanding the Structure of Job Listings

Most job listings are divided into several sections. Each section contains important information that helps us evaluate the opportunity before applying.

13.2.1 Overview of the Company and the Role

This section provides a broad description of the job position and the hiring company.

13.2.2 Responsibilities and Tasks

This section outlines:

- What the employee will be expected to do.
- What the employee will be responsible for.
- How the employee will function within the company.
- Which teams or individuals the employee will work with.

13.2.3 Requirements and Qualifications

This section describes what the company is looking for in a candidate. It may include:

- Educational background
- Certifications
- Years of experience
- Hard skills
- Soft skills

Some qualifications are listed as **required**, while others are **preferred** or **optional**.

Ideally, we should have:

- All the required qualifications.
- At least **50%** of the preferred qualifications.

If we identify requirements we are missing, we should consider whether we possess other skills or experience that may compensate for those gaps.

13.3 Full-Time (FTE) vs. Contract Positions

One important aspect to notice is whether a job is listed as **Full-Time (FTE)** or **Contract** (also referred to as **Contingent** or **Temporary**).

This refers to the **tax status of the worker**, not the number of working hours. Many contractors work full-time hours. However, FTE employees and contract employees often have different working conditions and contractual terms of employment. Both have advantages and disadvantages.

13.3.1 Characteristics of Full-Time (FTE) Employees

Full-time employees:

- Can expect to continue working indefinitely.
- May be paid more regularly, either through salary or hourly wages.
- Must comply with internal company policies, such as required training and work procedures.
- Are entitled to receive company benefits and incentives.

- Usually have different legal rights regarding termination and labor practices (although this may not be true in some states).

13.3.2 Characteristics of Contract (Contingent) Workers

Contract workers:

- Work for a period of time specified in their contracts.
- Receive payment either by hourly rate or by deliverables.
- May receive little or no training because they are expected to begin work with the necessary skills.
- Generally do not receive company benefits or incentives.
- Usually may be terminated without cause or notice and may not be covered by certain workplace protections.

13.4 Following Company Application Policies

If we conduct a job search using a website, we should carefully read the application instructions in every job posting.

[Each company has its own application process, and failing to follow the instructions may disqualify our application.](#)

Many companies advertise job openings on job-search websites to reach more candidates but require applicants to submit applications through their own company websites.

These application processes commonly require:

- Completing online application forms.
- Uploading supporting documents.

Some employers instead provide:

- An email address.
- A telephone number.

Applicants are then instructed to contact a specific person or team.

Depending on the employer, additional application requirements may include:

- Cover letter
- Work sample
- Portfolio link
- Screening questions
- Skill assessments

The key takeaway is to carefully read and understand the application instructions before submitting a job application.

13.5 Evaluating Whether a Company Is a Good Fit

Just as employers review applications to determine whether candidates are a good fit, we should also determine whether a company is a good fit before applying.

The following warning signs should be considered.

13.5.1 Lack of Clarity in the Job Posting

A job listing should clearly communicate the position.

Warning signs include:

- A job title that is too general.
- A vague or unclear job description.
- Unclear role boundaries.
- Excessive use of unnecessary jargon, acronyms, or unclear language.

The overuse of unclear language may indicate that the company does not communicate effectively with employees or applicants.

13.5.2 Unrealistic Expectations

Some job listings contain unrealistic expectations, which may suggest that:

- The company does not fully understand the position.
- The company may exploit new hires or set them up for failure.

Examples include:

- Entry-level jobs requiring several years of experience.
- Positions expecting high-level expertise across too many different areas.

13.5.3 Evident Biases or Prejudice

Companies should not include language that explicitly excludes or gives preference to candidates based on personal characteristics.

These characteristics include:

- Race
- Religion
- Gender
- Age
- Orientation
- Marital status
- Nationality
- Parental status
- Military status

- Appearance
- Abilities that are irrelevant to the job description

Such language is a significant warning flag and should be carefully considered before applying.

13.6 Key Takeaways

In this topic, we learned that:

- Job listings contain different sections that help us assess our qualifications.
- Full-time (FTE) and contract positions have different characteristics.
- Carefully following each company's application process improves our chances of success.
- Some job listings contain warning signs that may lead us to reconsider applying.

Every hiring company should aim to make a job posting clear and inviting, especially for qualified candidates.

The screening, selection, interviewing, and hiring processes should be executed intentionally and follow a transparent series of steps.

A company's job listings create its first impression, and they can also provide our first indication of the company's culture and competence.

14 Applying for a Job

Applying for a job involves more than simply submitting a resume. Every company has its own application process and requirements, so we should carefully follow the instructions and prepare our application materials to maximize our chances of success.

14.1 Applying for a Job

When we find a job listing for a role that interests us and for which we are well qualified, it is time to begin the application process. Since every company has its own procedures, we should carefully read the job listing and follow the instructions exactly as provided.

Companies may ask us to:

- Complete an online application form.
- Apply through a job platform such as LinkedIn.
- Send a resume and cover letter by email.
- Follow another specified application procedure.

Following the company's instructions precisely is essential. Failure to do so may result in our application being disqualified before it reaches the hiring managers.

There is generally no limit to the number of applications we can submit when applying to different companies. However, some companies restrict the number of applications that can be

submitted within a specific period. In such cases, we should be strategic and prioritize the positions that best match our qualifications and interests.

If multiple positions within the same company are appealing, one effective approach is to:

- Apply first for the position we are most qualified for or most interested in.
- Contact the Human Resources (HR) department or the hiring manager.
- Ask whether we should submit applications for multiple positions or avoid duplicate submissions.
- Determine whether the company can forward our application internally or allows us to create a profile for multiple applications.

14.2 Resume and Cover Letter

In most job applications, we will need to submit a resume, and in many cases, we should also include a cover letter even if it is not explicitly required.

If we already have basic versions of these documents, we only need to customize them for each specific position instead of creating new ones from scratch.

Our resume and cover letter should clearly demonstrate how our skills and experience match the requirements of the position.

14.2.1 Resume Checklist

Before submitting our resume, we should carefully check for the following:

- Outdated or missing contact information.
- Grammatical errors.
- Formatting errors.
- Using an unprofessional email address.
- Cliché vocabulary or unnecessary jargon.
- Unrelated job experience.
- Irrelevant hobbies.
- [Attaching a photo of ourselves \(our resume should not contain a photo\).](#)

14.2.2 Cover Letter Checklist

Before submitting our cover letter, we should ensure that it does not contain:

- Grammatical or formatting errors.
- Excessive length (multiple pages) or an overly brief message.
- An incorrect recipient or a misspelled name.
- Generic content that is not tailored to the company and job opening.
- An arrogant or overly humble tone (maintain a professional and confident tone).
- References listed in the letter.
- Explanations about why we left our previous employer or other issues that are better discussed during an interview if necessary.

14.3 Other Application Documents

In addition to a resume and cover letter, employers may request supporting documents to verify our qualifications.

These documents may include:

- Educational transcripts.
- Writing samples.
- Portfolios.
- Certifications.
- A list of references.
- Letters of recommendation.

The required supporting documents vary depending on the company and the position. We should always follow the company's instructions carefully to avoid submitting an incomplete application.

14.4 Document Naming and Organization

Before uploading PDF or other formatted documents, we should use a consistent file naming convention.

Include our full name in every file name.

Example:

- John_Smith-Resume.pdf
- John_Smith-Cover_Letter.pdf
- John_Smith-References.pdf
- John_Smith-Transcripts.pdf

Using consistent file names ensures that all submitted documents appear together when recruiters sort files alphabetically.

14.4.1 File Naming Best Practices

- Include our full name in every file.
- Use a consistent naming convention.
- Limit special characters to underscores (_) and hyphens (-).
- If spaces are used, use them consistently throughout all file names.
- Do not include version numbers in file names.
- Submit all requested documents at the same time whenever possible.

14.5 Creating a Cover Letter Template

Creating a reusable cover letter template can save time when applying for multiple positions.

Our template should include:

- Name.
- Job title.
- Contact information.

We can start with a previously written cover letter and identify the sections that need to be customized for different companies or job titles. Highlighting those sections makes future editing easier.

Save the template with the word "**template**" in the filename.

For example:

- [John_Smith-Cover_Letter-Template.pdf](#)

When applying for a specific position, replace "**template**" with the company's name or the job title before submitting the document.

14.6 LinkedIn Profile

LinkedIn is an essential platform for modern job seekers.

A large percentage of recruiters and hiring professionals use LinkedIn to search for candidates, making it an important part of the job application process.

With LinkedIn, we can:

- Search for job opportunities.
- Submit applications.
- Build our professional network.
- Share professional information.
- Increase our visibility with recruiters.

Our LinkedIn profile should be complete and consistent with the information provided in our resume and other application materials.

We should also:

- List our skills thoroughly to improve search visibility.
- Request endorsements from colleagues.

14.6.1 Tips for Using LinkedIn Effectively

- Track job applications using the **Jobs** homepage.
- View:
 - Jobs we applied for.
 - Application dates.
 - Whether our applications have been reviewed.
- Upload cover letters and attach them with resumes during applications.

- Cover letters are also scanned, increasing the chances of appearing in recruiter searches.
- Follow companies where we are applying.
- Follow prominent professionals in our field.
 - Recruiters can see our interests, and we may also receive valuable industry information.

14.7 Key Takeaways

- Always follow the company's application instructions exactly.
- Customize our resume and cover letter for every position.
- Carefully review all documents for grammar, formatting, and accuracy.
- Submit every requested supporting document.
- Use a consistent file naming convention that includes our full name.
- Maintain a complete and professional LinkedIn profile.
- Track our applications and continue networking through LinkedIn.

14.8 Conclusion

Our job application creates the first impression that an employer has of us as a candidate. We should always follow the company's instructions carefully and use every application as an opportunity to demonstrate our qualifications and career aspirations.

Accuracy, correctness, and professionalism across all supporting documents strengthen the image we present to recruiters and help us stand out from other candidates. A well-prepared application increases the likelihood that recruiters will want to learn more about our skills, experience, and how our career goals align with the company's mission and objectives.

15 Interview Rehearsal

15.1 Learning Objectives

After completing this topic, we will be able to:

- Explain why it is important to rehearse for job interviews.
- Describe the types of material to develop and prepare for an interview.
- Rehearse the verbal components of an interview effectively.

15.2 Introduction

While applying for jobs, we can also begin preparing for interviews. We do not need to have an interview scheduled before we start preparing. It is always helpful to complete the groundwork in advance so that when an interview is scheduled, we are ready.

One of the best ways to prepare is to rehearse what we will say and do. Many people neglect this step, but it can make a significant difference in creating a great impression.

If we are well prepared for an interview, we will have gathered a great deal of information. However, unlike submitting a research paper, we must present our knowledge verbally. There

is a significant difference between knowing information and being able to present it smoothly. We can bridge that gap by rehearsing our delivery and presentation.

Rehearsing is not only important for interviews but is also a valuable soft skill that we will use in almost any professional role.

15.3 Preparing an Elevator Pitch

An elevator pitch is a brief introduction of ourselves and our goals.

It should typically consist of **three or four sentences** and include:

- Our name and professional title.
- Our recent experience.
- What we are currently looking for in a job.

An elevator pitch is useful for networking and for answering questions such as:

- "Tell us about yourself."

15.3.1 Example Elevator Pitch

"My name is Angela Menendez and I'm a data scientist. For the past two years, I've worked in an entry-level role helping derive business insights from data using established models at an e-commerce company. I would eventually like to move into designing machine learning models, and I'm looking for a mid-level position with opportunities to work toward that goal."

Once we have composed our elevator pitch, we should:

- Memorize it.
- Practice saying it repeatedly until it feels natural.
- Become comfortable delivering it whenever needed.

The easier it is to present our elevator pitch naturally, the more useful it becomes.

15.4 Preparing Answers to Common Interview Questions

Every interview includes unique questions, but many interviewers ask similar questions.

Some common interview questions include:

- Why did you choose this field, and why are you interested in this job?
- Tell us about a time you solved a problem on a project and how you did it.
- Where do you see yourself in five years?

We can find discussions of common interview questions on many websites. After researching them, we should prepare our own original answers.

Our resume is also a valuable resource because it helps trigger memories about:

- Skills we have used.
- Projects we have worked on.
- Tasks we have completed.

15.5 Preparing Questions for the Interviewer

Most interviews include an opportunity for us to ask questions.

Instead of relying on whatever comes to mind during the interview, we should prepare questions in advance.

Interviewing is a two-way process, so we should prepare thoughtful questions.

Some useful approaches include:

- Use notes from researching the company and the interviewers to customize our questions.
- Ask about the company's products.
- Ask about the company's organizational structure.
- Ask about recent news releases.
- Ask about topics related to our career plans and preferred work style.
- Ask whether the team works closely together or independently.
- Ask about the promotion path from the role.

After preparing these questions, we should practice asking them aloud. The more comfortable we become asking them, the easier it will be to focus on the interviewer's answers and respond naturally.

15.6 Rehearsing the Interview

Once we have prepared:

- Our elevator pitch.
- Our interview answers.
- Our questions for the interviewer.

We should rehearse all of them.

A good rehearsal process includes:

- Reading our responses aloud.
- Paraphrasing our responses rather than reading them word-for-word.
- Practicing until we are comfortable speaking from our notes instead of memorizing every sentence.

It is not necessary to memorize everything. Instead, we should practice until speaking naturally becomes comfortable.

15.7 Improving Our Delivery

We can further improve our interview performance by using the following techniques:

- Practice in front of a mirror to identify nervous habits.
- Learn to maintain eye contact.
- Record ourselves and watch or listen to evaluate how we appear and sound to others.

Rehearsing is like a secret weapon during the interview process. Many candidates skip this step, so practicing provides a real advantage by increasing our confidence and poise.

15.8 Practicing Through Mock Interviews

When we feel comfortable with our preparation, we should ask a friend to conduct a mock interview.

During the mock interview, our friend can:

- Ask common interview questions.
- Help us practice verbalizing our answers naturally.
- Help us become more comfortable answering questions under the pressure of a real interview.

If we feel comfortable, we can also ask our friend to:

- Ask questions that we have not researched beforehand.
- Help us practice thinking of answers quickly.
- Allow us to practice asking the questions we plan to ask during the actual interview so we become comfortable with their wording.

15.9 Key Takeaways

- Rehearsing before an interview helps us become comfortable presenting ourselves.
- Practicing an elevator pitch is an important part of interview preparation.
- We should prepare answers to common interview questions.
- We should also prepare thoughtful questions to ask the interviewer.
- Effective rehearsal techniques include:
 - Practicing aloud.
 - Speaking in front of a mirror.
 - Recording ourselves.
 - Participating in mock interviews.
- Taking time to rehearse and prepare is well worth the effort because it reduces interview stress and helps us stand out from other candidates.

Module 3: Interviewing

16 Overview of the Interview Process

16.1 Learning Objectives

After studying this topic, we will be able to:

- Identify the factors that affect the interview process.
- List some of the common steps of an interview process.
- Describe what we can expect during the different stages of interviewing.

16.2 Factors That Affect the Interview Process

Each job interview is unique. Several factors can make the interview process very different, even within the same company. These factors include:

- The size of the company and the company culture.
- The size of the working team or department.
- The number of positions to be filled.
- The type of position.

An interview at a large company will often be very different from an interview at a small company. Similarly, the interview process for an entry-level position will be different from that of a lead or higher-level position.

It is helpful to remember these differences when we try to anticipate what to expect during an interview.

16.3 Common Interview Process

After we apply and are accepted as a candidate for a position, job interviews typically follow a common pattern of steps.

16.3.1 Technical Assessment

If a company is interested in a candidate for a technical position, they will often give a code challenge or technical test to complete.

Code challenges for Data Scientist candidates usually involve applying both SQL and Python to solve Machine Learning problems.

This is the first opportunity for candidates to demonstrate their technical skills. For a technical role, it is crucial for a company to ensure that a candidate has the necessary abilities to perform the job. Usually, candidates must perform well in these technical assessments to move forward in the interview process.

16.3.2 Interview Rounds

The next stage of the interview process involves different types of interviews. Candidates may complete any number of these interview rounds.

- **Team Interview**
 - Candidates may speak with a team leader or project manager.
 - They may also meet other team members.
 - The purpose is to determine whether the candidate can work well with the team and understands the role.
- **Human Resources (HR) Interview**
 - HR interviews usually focus on the company's definition of the role.
 - They may also cover company policies and other employment-related information.
- **Technical Interview**
 - This interview is often an expansion of the initial technical assessment.
 - Candidates discuss their technical skills in greater depth.
 - They may also discuss the technical aspects of the job and demonstrate their problem-solving abilities.

Candidates may complete any number of these interview rounds depending on the company's hiring process.

16.3.3 Final Interview

If a candidate performs well in the previous stages, they may be invited to a final interview.

During the final interview:

- Candidates may meet with higher-level managers.
- They will likely discuss the role in greater depth and detail.
- They may be asked more situational and behavioral questions.
- They have one final opportunity to ask questions that clarify the role and help them better understand the company.

After the final interview, the company decides whether or not to extend a job offer.

16.4 Key Takeaways

In this topic, we learned that:

- Many factors can affect the interview process.
- Many interview processes include first-round, second-round, and final interviews.
- Each interview round has predictable features that we can prepare for.
- Every employer has its own way of conducting job interviews.
- Understanding the interview process and preparing for each stage helps us make the best possible impression.

17 Expert Viewpoints: A Typical Interview Cycle

17.1 Overview

In this topic, experts describe a typical interview cycle for data science candidates and provide practical advice on preparing for interviews, making a strong first impression, and demonstrating the qualities that interviewers look for beyond technical skills.

17.2 A Typical Data Science Interview Cycle

A data science interview is often casual in its structure, but companies commonly require candidates to complete one or more technical assessments.

Typical interview tasks may include:

- Solving a programming or analytical problem within **one hour**.
- Answering a set of technical questions.
- Completing a **live coding** session (writing or debugging code in real time).
- Fixing an existing program and finding bugs (errors in the program).

We should be prepared for these types of interview activities before attending the interview.

17.3 Making a Good First Impression

Interviewers place significant importance on the first impression we create. A positive first impression is influenced by our energy, attitude, communication style, and body language.

17.3.1 Show Enthusiasm and High Energy

When we enter an interview, we should:

- Be energetic, even if the interviewers appear less energetic.
- Demonstrate genuine interest in the role.
- Show through our energy level that we are excited about the opportunity.

17.4 Personal Qualities Interviewers Look For

In addition to technical skills, interviewers evaluate several non-technical attributes.

They look for:

- The effort we put into solving a problem.
- Our approach to finding a solution.
- Our ability to think outside the box.
- Our overall attitude toward the interview.
- Whether we are excited about the opportunity.
- Whether we are coachable (willing to accept guidance and improve based on feedback).

During discussions, interviewers may explore multiple solutions or take the conversation in different directions ("go down a rabbit hole" (**explore one line of thinking in great depth before returning to the main discussion**)). They observe how well we respond to feedback and adapt our thinking.

17.5 Answering Questions When We Do Not Know the Answer

Candidates should avoid simply saying:

"I don't know."

Saying only "I don't know" can sound as though we have given up.

Instead, we should:

- Explain the reasoning or context behind why we do not know the answer.
- Mention whether we have not worked in that area before.
- Explain how we would approach solving the problem.
- Demonstrate our problem-solving process, even if we cannot immediately recall the correct answer.

This helps the interviewer understand that although we may not know the answer immediately, we have a structured way of solving unfamiliar problems.

17.6 Tips for Live Coding Interviews

During live coding sessions:

- Never become flustered (overly nervous or confused) when something goes wrong.
- Remember that coding mistakes happen in the real world.
- Even small syntax errors, such as missing or misplaced **colons** (:), can occur.
- Think aloud while solving the problem.
- Explain our reasoning throughout the coding process.
- Pay attention to hints that the interviewer may provide.

Talking through our thought process allows the interviewer to understand how we solve problems, even if the final solution is not immediately correct.

17.7 Body Language and Communication

Since interviews are largely about making a positive first impression, we should be aware of both our verbal and non-verbal communication.

Avoid distracting behaviors such as:

- Tapping our foot.
- Making unnecessary repetitive movements.

Instead, pay attention to:

- Our tone of voice.
- Our hand movements.
- Our overall body language.
- Clear and confident communication.

Positive body language helps create a stronger impression during the interview.

17.8 Resume and LinkedIn Preparation

Whenever we apply for a job, we should take the application process seriously.

Before the interview:

- Thoroughly understand our resume.
- Ensure our **LinkedIn profile** and resume are aligned.
- Review how our resume closely matches the job description.
- Be prepared to explain every item listed on the resume.
- Tailor our preparation according to the specific job requirements.

17.9 Demonstrating Professionalism

If we are serious about getting the job, we should demonstrate:

- A positive attitude.
- Positive body language.
- Excellent communication skills.
- Confidence in presenting our qualifications.
- The ability to effectively market ourselves for the job opportunity.

17.10 Final Interview Advice

Experts recommend the following:

- Relax.
- Be yourself.
- Be brave.
- Do not act or pretend to be someone else.
- Be honest throughout the interview.

If we genuinely do not know an answer:

- Clearly say that we do not know.
- Do not waste time trying to bluff or "hand wave" (give vague or unsupported explanations instead of a clear answer).
- Move on to the next question if appropriate.

If we have a reasonable idea, we can respond by saying:

"I don't know, but I guess..."

or

"I would solve this in the following way..."

Explaining our thought process is often considered a good approach because it demonstrates logical thinking and problem-solving ability, even when we do not know the exact answer.

18 Unethical Interviewing

As candidates, we work hard to earn interview opportunities. We want to cooperate with the hiring company and secure the job. However, not every interviewer has our best interests as a priority. It is important to understand which interview questions are inappropriate or off-limits so that we can protect our rights and recognize whether a company demonstrates ethical and respectful hiring practices.

Note: Laws and hiring practices differ across countries and regions. This topic covers these issues in a general sense. We should always research the employment laws and interview regulations applicable to our own location.

18.1 Personal Information

There are several categories of personal information that interviewers generally should not ask about.

18.1.1 Race or Nationality

- Asking about race or nationality is strictly illegal in the United States and many other countries.
- Some interviewers may attempt to obtain this information indirectly by asking about:
 - The origin of our name.
 - Our first language.
- Some questions related to citizenship status and work visas may be permitted because they are directly related to employment eligibility.
- For example:
 - Asking whether we are **legally allowed to work in the country** is generally permissible.
 - Asking directly whether we are a citizen is generally not appropriate.

18.1.2 Gender and Sexual Preference

- In the United States, discrimination based on gender or sexual preference is illegal.
- Therefore, interviewers should not ask questions about these topics.

18.1.3 Marital Status and Family Plans

- Asking about marital status is illegal in the United States.

- This information is generally unrelated to job performance.
- Such questions may be used to determine whether a candidate will be as dedicated as the company expects.
- Questions about family plans may be intended to determine whether female employees might require future leave or expensive insurance coverage.
- It is also unethical for employers to ask for information about our spouse because it invades personal privacy.

18.1.4 Health History

This includes our personal health history, family health history, and medical background.

- Unless there is a specific job-related reason (which is rare), interviewers should not ask about these topics.
- Such questions may be used to estimate future insurance costs or sick leave requirements, which is unethical.

18.1.5 Disability

- If a job has specific physical requirements, an interviewer may ask whether we are able to perform the required job duties.
- In the United States, employers must provide **reasonable accommodations** (changes or support that enable qualified individuals with disabilities to perform their jobs) for disabled employees.
- It is also unethical for employers to ask:
 - Whether we have ever been injured at work.
 - Whether we have ever received workers' compensation.

18.1.6 Age

- Employers may ask whether we meet the legal age requirement for a position when required by law.
- General questions about age that are unrelated to the position are unethical.
- Interviewers may also try to estimate our age indirectly, such as by asking when we graduated.

18.1.7 Legal or Arrest Record

- Interviewers should not ask about our legal history or arrest record.
- If an employer has a legitimate business reason to obtain this information, official legal channels are available.

18.1.8 Financial History

- Some employers may perform a credit check after obtaining our permission during the hiring process.
- However, interviewers should not ask questions about:
 - Credit history.
 - Bank accounts.

- Bankruptcy history.
 - Other personal financial matters.
- Be aware of indirect questions, such as those about home ownership.

18.2 Information About Previous Employers and Employment

Interviewers may already know some information from our resume, references, and employment history. However, certain topics remain inappropriate.

18.2.1 Company Information

- Interviewers may ask about:
 - Our projects.
 - Our responsibilities.
 - Our job tasks.
- They should not ask for confidential internal company information, such as:
 - Company structure.
 - Internal projects.
 - Proprietary information.
- Sharing such information may violate previously signed **Non-Disclosure Agreements (NDAs)** (legal agreements that prohibit sharing confidential information).

18.2.2 Coworkers and Managers

- Employers may contact references that we voluntarily provide.
- However, interviewers should not ask us for confidential or inside information about coworkers or managers from previous employers.

18.2.3 Compensation

- Until we reach the salary negotiation stage with a concrete job offer, we should be cautious when discussing previous salaries.
- Instead, we should focus on:
 - The employer's salary range.
 - The typical market salary for similar positions in our area.
- Sharing previous salary information may:
 - Continue unfair compensation practices.
 - Encourage employers to offer less than the fair market salary.

18.3 Personal Behavior

Interviewers should generally avoid questions about our personal behavior unless it directly affects our ability to perform the job.

18.3.1 Religion

- In the United States, interviewers may not ask about our religious beliefs or affiliations.
- Be cautious of indirect questions, such as:

- Where we worship.
- How we spend weekends.
- Whether we are available to work weekends.

18.3.2 Testing

- Some positions legally require specific tests, such as a urine test.
- Employers may inform us about required testing after a job offer has been accepted and obtain our permission.
- It is unethical to ask during the interview whether we would agree to such testing and make hiring decisions based on our answer.
- The same principle applies to:
 - Lie detector tests.
 - Genetic tests.
 - Similar forms of testing.

18.3.3 Use of Substances

Interviewers should not ask whether we:

- Smoke.
- Drink alcohol.
- Take medications.
- Participate in other substance-related activities during our personal time.

If drug testing is required for the position, employers should notify us after extending a job offer and perform the testing only with our permission.

18.3.4 Political Opinions

- Interviewers should not ask about:
 - Our political beliefs.
 - Political affiliations.
 - Opinions about labor unions.

18.3.5 Personal Appearance

- Employers may discuss workplace dress standards if they are relevant to the position.
- However, interviewers should not comment on or question our personal appearance.
- Such questions may be used to infer:
 - Socio-economic status.
 - Gender.
 - Religious practices.
 - Race (for example, hairstyles).

18.3.6 Culture Fit

- Some organizations evaluate whether candidates fit their workplace culture.

- We should be cautious because the term "**culture fit**" may sometimes be used to disguise discriminatory hiring preferences by favoring people who resemble those already employed (for example, candidates who are white, male, younger, or American).

18.4 Conclusion

In most cases, employers are ethical, follow employment laws, and strive to treat candidates fairly. However, some interviewers may ask inappropriate questions.

If this happens, we have several options:

- Deflect the question by giving an answer that relates more appropriately to the position or our qualifications.
- Politely state that we do not wish to answer the question.
- Continue the interview without discussing inappropriate personal information.

If refusing to answer makes us a less desirable candidate, we should consider whether we truly want to work for an organization that demonstrates questionable hiring practices or unethical interview behavior.

19 Interview Experience and Advice from a Hiring Manager and a Successful Candidate

19.1 Overview

In this topic, Antonio and Cindy review their experiences during the interview process and share practical advice for applying to data science positions, particularly internships. They discuss resume preparation, networking, LinkedIn, job searching, and how candidates can improve their chances of standing out in a competitive hiring process.

19.2 The Hiring Process for Data Science Positions

Whenever there is a data scientist position, especially for interns, companies can receive **up to 1,000 applications**. In many cases, the application form must be closed early to prevent receiving **5,000 or more candidates**, making it impossible to review every application.

The hiring process typically involves:

- Receiving up to **1,000 applications**.
- Narrowing them down to approximately **100 resumes**.
- Scheduling interviews with shortlisted candidates.
- Selecting the final **3 to 5 candidates**, depending on the number of available positions.

Because of the large number of applicants, every part of the application must help us stand out.

19.3 Tailoring Our Resume for Every Job

One of the reasons Cindy was selected was that she had an excellent resume specifically tailored for the position.

Instead of using the same resume for every company, we should:

- Carefully review the job requirements.
- Highlight only the experiences that are relevant to that specific position.
- Showcase projects, skills, and accomplishments that directly match the employer's needs.

Important Advice:

- Never use the same resume repeatedly for different companies.
- Always customize the resume according to each job posting.

19.3.1 A Practical Resume Strategy

One effective technique is to maintain a separate document containing all of our:

- Experiences
- Skills
- Projects
- Achievements

Then, based on each job posting, we can select the most relevant experiences and projects to include in the resume, maximizing our chances of receiving a response from the company.

19.4 Writing a Professional Cover Letter

A strong cover letter can create an excellent first impression.

Cindy's cover letter was:

- Friendly
- Professional
- Well-written

It immediately gave the hiring manager the impression that she would be an excellent candidate for the position.

19.5 Using LinkedIn to Gain an Advantage

In addition to submitting her application, Cindy also:

- Connected with the hiring manager on LinkedIn.
- Sent a professional message.
- Asked thoughtful questions.

- Expressed genuine interest in the position.

These actions created the impression that she was:

- Highly interested in the role.
- Motivated to begin working.
- A strong fit for the position.

Although contacting the hiring manager is **not a requirement**, it can provide a valuable advantage during the hiring process.

Many successful data scientists have been hired without contacting recruiters on LinkedIn. However, reaching out professionally can help us become more memorable.

19.6 Why Contacting Recruiters Can Help

When hundreds of resumes are submitted for a single position, it is difficult for any individual resume to stand out.

To improve our visibility, we should consider:

- Connecting with the hiring manager.
- Connecting with the recruiter.
- Sending a professional message.
- Asking relevant questions about the position.
- Expressing our genuine interest in joining the company.

These actions help recruiters remember our names before reviewing our resume.

Later, while reviewing applications, the hiring manager may recognize our name and naturally pay a little more attention to our resume.

Although this does not guarantee an interview, it can provide a small but meaningful advantage.

19.7 The Importance of Networking

One important piece of advice is that **networking is extremely valuable**.

Some people may feel:

- Shy about contacting professionals.
- That industry professionals do not want to talk to people they do not know.

However, this is generally not true.

Many professionals in the industry are willing to:

- Share their expertise.
- Answer questions.

- Offer career advice.
- Discuss their experiences.

If we demonstrate genuine passion and interest, they may even:

- Refer us for a job.
- Inform us about job openings within their company.

19.8 Learning Beyond University Education

Another important lesson is that we should not limit our learning to university education.

Universities teach many valuable theoretical concepts, but many courses do not always teach us how to apply those concepts in real-world industry settings.

Therefore, we should continue learning independently by:

- Building projects.
- Practicing technical skills.
- Learning industry tools.
- Applying theoretical concepts to real-world problems.

19.9 Why LinkedIn Connections Matter

When hiring managers have already interacted with us on LinkedIn, they become familiar with our names and our interest in the position.

If they have already:

- Answered our questions.
- Seen our enthusiasm.
- Learned about our commitment to the role.

Then, when they later review our resume, our name is more likely to be recognized.

This recognition can encourage them to spend a little more time reviewing our application.

Again, this is only a small advantage, not a guarantee of being selected.

19.10 Job Hunting Is About More Than Technical Knowledge

Finding a job is not only about what we know.

It is also about who we know.

Building professional relationships allows us to:

- Learn more about specific roles.
- Leave a positive impression.

- Increase our chances of receiving an interview.
- Give recruiters an opportunity to know us beyond our resume.

19.11 Cindy's Job Search Experience

During a period of **three to four months**, Cindy:

- Applied to **approximately 100 or more positions**.
- Received **around five interviews**.
- Fortunately, IBM was one of those interview opportunities.

Although data science is currently in high demand, there is also significant competition from many qualified candidates.

19.12 Why LinkedIn Is an Excellent Platform for Job Searching

LinkedIn was one of the primary platforms used during the job search because it provides:

- A large number of job postings.
- Updates from data science and technology influencers.
- Career advice from industry professionals.
- Information about available positions shared by companies and professionals.

This makes LinkedIn valuable not only for finding jobs but also for staying informed about developments within the data science and technology industries.

19.13 Final Advice for Job Applicants

To improve our chances of securing interviews, we should:

- Tailor every resume to the specific job.
- Maintain a master document containing all experiences and skills.
- Write a professional and friendly cover letter.
- Connect with recruiters and hiring managers on LinkedIn.
- Ask thoughtful questions about the position.
- Express genuine interest in the role.
- Build a strong professional network.
- Continue learning beyond university coursework.
- Gain practical experience by implementing theoretical concepts in real-world projects.
- Use LinkedIn actively for networking, learning, and discovering job opportunities.

These strategies may not guarantee a job offer, but together they can significantly improve our ability to stand out among hundreds of applicants and increase our chances of receiving interview opportunities.

20 Preparing for a Data Science Interview

20.1 Interview Preparation Strategies

Proper preparation plays an important role in performing well during an interview. Effective preparation includes reviewing the job requirements, refreshing technical knowledge, and practicing interview responses.

One interviewee explained that they prepared for the interview the same way they would prepare for any other interview. The main preparation steps included:

- Reviewing the job description to understand the required knowledge and skills.
- Looking up any topics or concepts they were unsure about.
- Going through their resume and ensuring they could explain every point in thorough detail.
- Refreshing previously learned technical skills, such as programming languages that had not been used for some time.
- Relying on their previous experience and background knowledge, which made it easier to answer technical questions.
- Practicing practical programming questions before the interview.

Another interviewee described spending a significant amount of time creating a personal cheat sheet that contained:

- Fundamental concepts related to Data Science and Machine Learning.
- Common interview questions.

Before each interview, they reviewed this cheat sheet and practiced explaining every concept either with or without looking at the answers. They even explained the concepts aloud while standing in front of a mirror to improve their communication skills.

20.2 Practicing Interview Responses

One of the most important interview preparation strategies is practicing answers before the actual interview.

An interviewee explained that they regularly practiced answering interview questions while looking into a mirror. Later, they wanted to evaluate their performance more carefully, so they began recording themselves using their phone while answering questions.

Reviewing these recordings helped them identify several areas for improvement, including:

- Poor posture.
- Speaking too quickly when feeling nervous.
- Stuttering while answering questions.
- Overall presentation and communication style.

By reviewing these recordings, they were able to refine these behaviors and present themselves more confidently during interviews.

20.3 Handling Technical Questions During an Interview

According to the interviewer, it is completely normal for candidates not to know the answer to every technical question.

Technical questions are generally tailored to the candidate's level of technical knowledge.

- If a candidate demonstrates limited technical knowledge during the early part of the interview, easier questions are asked.
- If a candidate consistently provides excellent, textbook-quality answers, the interviewer may ask much more challenging questions to determine the limits of the candidate's knowledge.

Being asked difficult questions does **not** mean that the interviewer expects every answer to be correct. Instead, it often indicates that the interviewer recognizes the candidate as highly knowledgeable and wants to evaluate the depth of their understanding.

As a result, almost every candidate will eventually encounter questions for which they do not know the exact answer.

20.4 Responding When We Do Not Know the Exact Answer

When we are unsure of an answer, the best approach is to demonstrate our reasoning ability rather than pretending to know the answer.

A recommended response is:

"I don't know for sure," or "I don't know the answer, but I'm going to take an educated guess."

After saying this, we should explain the reasoning behind our educated guess.

An example provided during the interview was:

"My educated guess would be that the cost of linear regression will be lower because you don't always have to consider the penalty term. Sometimes that penalty term can cause the function to keep on training until it finally reaches a point where there's no more penalty, whereas linear regression does not need to consider that, so it could potentially be faster."

Even if the answer is not completely correct, demonstrating logical reasoning and having sufficient background knowledge can earn positive recognition from the interviewer or hiring manager.

However, if we are completely unfamiliar with a topic and lack the fundamental knowledge required to answer the question, we should avoid pretending to know the answer.

Instead, it is better to respond honestly by saying:

"I'm sorry, I'm not quite familiar with that."

or

"I'm sorry, I don't quite know the answer to that."

Honesty creates a better impression than confidently providing an incorrect answer.

20.5 Arriving for the Interview

Punctuality creates a positive first impression.

We should:

- Arrive at least **10 to 15 minutes early**.
- Plan the journey in advance because unexpected delays may occur, such as a delayed bus or other transportation issues.
- Remember that arriving early always creates a better impression than arriving late.

20.6 Dressing for the Interview

Appropriate clothing depends on the company's culture.

One interviewee explained that, for the role at **Skills Network Labs**, they got the impression that the company culture was professional while also being enjoyable and friendly—not overly formal, but not goofy either.

Regardless of the company's culture, the priority should always be to look clean and well-presented.

21 Coding Challenges in Data Science

21.1 Learning Objectives

After completing this topic, we will be able to:

- Describe what a coding challenge or take-home quiz is.
- Summarize the process of taking a coding challenge or take-home quiz.
- Recognize a typical coding challenge problem.

21.2 Introduction

If an interview process includes a coding challenge or take-home questions, it is likely to be one of the most important steps in the hiring process.

In a coding challenge, we work on a problem independently and submit the completed solution to the interviewers for review. The coding challenge is intended to showcase our problem-solving and coding skills in a concrete way.

The company will usually send us files to work with or provide a URL where we can solve the problem online.

A coding challenge usually occurs after the initial screening phase, although it is sometimes the first step in the interview process.

21.3 Coding Challenges for Data Science Roles

For many data scientist roles, there may be both coding problems and technical take-home quizzes.

A coding challenge for a data science candidate is generally a data science problem that may include topics such as:

- Understanding the types and roles of variables in a dataset.
- Calculating descriptive and exploratory statistics to understand the variables in a dataset.
- Identifying the machine learning model that can be applied to the problem.
- Organizing and reporting results by presenting findings and including visualizations where appropriate.

Coding projects for data scientists vary in scope and complexity based on seniority and the job description.

21.3.1 Entry-Level Positions

For lower-level positions, the test may be relatively simple, such as exploratory data analysis (EDA).

Candidates should know:

- Basic data science skills.
- Common vocabulary and procedures.
- How to produce summary statistics.
- How to create charts and visualizations.

21.3.2 Higher-Level Positions

Higher-level projects may include more complex tasks, such as:

- Building a regression model.
- Building a classification model.
- Forecasting using a time-dependent dataset.

21.4 Types of Coding Challenges

Sometimes the coding challenge is directed, and the requirements clearly specify:

- Where to download the data.

- Whether the data should be preprocessed.
- What type of exploratory data analysis should be performed.

In this type of test, we should carefully follow all directions correctly.

For more experienced positions, the problem may be explained without providing clear guidance. In such cases, we need to:

- Determine suitable parameters for addressing the project goals and objectives.
- Explain the assumptions we made.
- Justify why those assumptions were chosen.
- Describe how we approached and solved the problem.

21.5 Datasets and Project Instructions

The interview team will provide project directions along with a dataset.

Sometimes, they may provide:

- A small, clean dataset.
- Data stored in a Comma-Separated Values (CSV) file format.

In other situations, the project directions will specify where to obtain the dataset.

The size and condition of the dataset may vary according to the position level:

- Entry-level challenges often use smaller, cleaner datasets.
- Higher-level challenges may involve large datasets that require cleaning or preprocessing.

Instructions for completing the challenge are usually provided.

The interview team should clearly specify:

- Project deliverables.
- Project deadline.

If any instructions are unclear, we should ask questions for clarification.

21.6 Take-Home Quiz vs. Coding Challenge

21.6.1 Take-Home Quiz

A take-home quiz typically has:

- A short deadline, often about **40 minutes**.
- An online format.
- No option to pause or exit during the quiz.
- No option to leave the page during the assessment.

Therefore, we should plan sufficient uninterrupted time before starting.

21.6.2 Coding Challenge

A coding challenge usually has:

- A firm submission deadline.
- More time to complete than a take-home quiz.
- A completion window ranging from **several hours up to a few days**.

21.7 Confidentiality and Ownership

When a company begins hiring for a position, it designs or sources a coding challenge based on the job description.

Companies expect candidates **not to share** the challenge or its solutions with other job seekers.

They may also have policies defining who legally owns the work completed during the coding challenge.

These requirements are often included in a **Non-Disclosure Agreement (NDA)** that candidates are asked to sign.

21.8 Sample Coding Challenge

A coding challenge always begins with instructions.

21.8.1 Instructions

For this sample challenge:

- Python should be used to solve the coding challenge.
- Python libraries may be used to solve the challenge.

After the instructions, a dataset is provided through a downloadable link.

The problem statement is then introduced.

In this coding challenge, the candidate is required to build a **regressor** (a machine learning model used to predict continuous numerical values) that recommends **crew size for potential ship buyers**.

After the problem statement, the required deliverables are described.

21.9 Required Deliverables

The challenge requires completing the following tasks:

- Read the file into a DataFrame and display all columns.
- Calculate the basic statistics for the variables **NOX**, **age**, and **tax**, including:
 - Count
 - Mean
 - Median
 - Standard deviation
- Examine the data and determine whether these variables contain outliers.
- Select the columns that are likely to be important for predicting the **median sale values of homes**.
- Drop the columns that are not required while clearly explaining the reason.
- Apply one-hot encoding to categorical features.
- Divide the dataset into training and testing sets.
- Use **60%** of the data for training and the remaining **40%** for testing.
- Select an appropriate machine learning model and explain why it was chosen.
- Build the machine learning model.
- Calculate the Pearson correlation coefficient for both the training and testing datasets.

21.10 Common Follow-Up Interview Questions

After completing the coding challenge, we may be asked questions such as:

- How would you improve the accuracy of your model?
- Can you describe the process of tuning the hyperparameters?
- What is the regularization parameter in your model?
- Does your model have a bias or variance problem?
- Plot the regularization parameter value versus the Pearson correlation coefficient for both the training and testing datasets to evaluate bias and variance.

21.11 Difficulty Level

This is a relatively simple coding challenge.

Problems of this type are generally designed for **entry-level data scientist positions**.

The instructions are clear and concise, so we simply need to:

- Follow the instructions carefully.
- Write the required code.
- Complete all required deliverables.

The typical time allowed for this type of coding challenge is **24 to 72 hours**.

21.12 Key Takeaways

- Coding challenges and take-home quizzes are often an important part of the interview process.
- We are generally provided with clear instructions and a specific time limit.
- Coding challenge problems become more complex depending on the level and nature of the role.

- Take-home quizzes and coding exercises provide an excellent opportunity to demonstrate our ability to complete a data science project.
- These assessments help the hiring team evaluate our technical and problem-solving skills.
- By taking our time, following the instructions carefully, and doing our best, coding challenges can become a valuable asset in our job search.

22 Expert Viewpoints: Case Study Insights

22.1 Understanding and Solving a Case Study

When approaching a case study, we should first understand **what business problem we are trying to understand or study** and identify the **hypothesis of the problem**. We should prepare for the problem in advance and ask a lot of good questions to the business stakeholders because the problem we are solving is ultimately for them to use.

Before beginning the solution, we should have the **end result in mind** and understand:

- What is the case study looking for?
- What is the target question we are addressing?
- What methodology should we use to tackle the problem?

If the case study involves a **machine learning problem**, we should:

- Understand how to construct the dataset.
- Follow the data transformation process.
- Build machine learning models.
- Train and deploy the machine learning models.
- Track and monitor the machine learning models after they have been trained, validated, and pushed to production to evaluate how well they perform.

Applying these steps to the business problem and the case study can be very helpful in developing a structured and effective solution.

22.2 Demonstrating the Right Mindset

Case study questions are very useful for determining whether a candidate has the right mindset and how they reason about problems.

When answering a case study question, we should:

- Take our time.
- Reason through the problem carefully.
- Avoid rushing to give the first answer that comes to mind.
- Focus on fully understanding what the interviewer is asking.

It is important to ask follow-up questions to clarify the case study. Asking clarification questions is usually a strong indication that we are a good candidate because it demonstrates that we want to fully understand the requirements before attempting to solve the problem.

In a real job, there will often be requirements that need clarification. Asking thoughtful questions shows that we are prepared to take on the work rather than simply accepting the task without fully understanding it.

Only after clarifying the situation should we begin answering the case study question.

22.3 Types of Case Study Questions

Case study questions are common in interviews and are typically business-specific.

For example, an interviewer might present a scenario such as:

- A client says that something we delivered was incorrect.
- A client says the work was delivered past the deadline.

The interviewer will then ask how we would handle that situation.

These questions may be:

- Very specific.
- Very ambiguous.

Both types are valuable because they help the interviewer evaluate our problem-solving approach.

22.4 What Interviewers Evaluate

When interviewers ask a case study question, they are primarily looking to determine whether we can:

- Apply logical reasoning.
- Make reasonable assumptions.
- Follow the correct steps based on those assumptions to solve the problem.

The emphasis is not only on arriving at the final answer but also on the reasoning process used to reach that answer.

22.5 How to Answer a Case Study Question

A highly effective approach is to **think out loud** throughout the entire problem-solving process.

We should:

- Make our assumptions verbally based on the information provided in the case study.
- Clearly explain our reasoning, such as:
 - "Since this is true, then we can assume this."
- State that these assumptions are based on the information we have been given.
- Continue building the solution step by step based on those assumptions.

- Explain where each assumption leads and how it influences the next step.

Walking through the problem logically while openly explaining our assumptions is one of the best approaches when answering a case study question because it allows the interviewer to evaluate our reasoning and problem-solving process.

23 Second-Round Screen

Receiving an invitation to a second job interview is a significant achievement. It indicates that the interviewers from the first round believe we may be a strong fit for the position.

Second-round interviews usually require us to provide more detailed information about our competencies, ethics, and professional outlook. Employers also want to determine whether we will fit well within the company's culture. Our answers should align with the company's mission and the responsibilities of the role. Since companies typically have open positions to support growth or improve existing processes, we should clearly demonstrate how we can contribute to achieving their goals.

23.1 General Second-Round Preparation

To prepare effectively for a second-round interview, we should reflect on the first interview and use that experience to improve our performance.

Consider the following:

- Reflect on what went well and use those strengths to our advantage.
- Make note of the topics that received the most attention or seemed especially important to the interviewers. Prepare follow-up questions related to those topics.
- Think about whether we would answer any previous question differently. If a question caught us off guard, consider how we would respond if given another opportunity.
- Recall the names and job positions of the people we met during the first interview.
- Write down any thoughts, questions, or concerns that arose after the first interview so they can be discussed if an opportunity arises.

23.2 Types of Second-Round Interviews

Second-round interviews are generally more focused than first-round interviews but less comprehensive than final interviews.

We may:

- Meet with the same interviewers from the first round.
- Meet with team leaders.
- Meet with technical managers.
- Meet with Human Resources (HR) personnel.
- Meet with executive or senior management.

Each company follows its own interview process.

23.3 HR Screen

An interview conducted by a Human Resources (HR) professional usually focuses on our previous experience, understanding of the company, and compatibility with its culture. Sometimes this interview may even serve as the first interview.

Generally, an HR screen does not involve detailed technical questions or in-depth discussions about the role.

Common questions include:

- What motivated you to apply for this job?
- What attracted you to our job listing?
- Why did you leave your last job?
- What is your ideal working environment?
- What do you know about our company's products or services?
- What are your expectations regarding salary and other compensation?

23.3.1 Tips for HR Interview Questions

When answering these questions:

- Relate your previous experience to the role.
- Maintain a positive and professional attitude.
- Prepare thoughtful responses to potentially uncomfortable questions.
- If we left a previous position for negative reasons, explain the situation in a neutral and professional manner.
- Avoid discussing our specific compensation history unless required.
- Be honest about our expected salary and explain how flexible we are regarding compensation.

23.4 Team Leader Interview

When interviewing with a team leader (often a project manager or another manager), the discussion typically focuses on the team, project workflows, collaboration, and work style.

Although some technical discussion may occur, behavioral questions are usually more important.

Common questions include:

- Tell me about a few of the first things you would do in this role.
- Give a few examples of best practices in data science.
- What's a data science project you would want to work on at our company?
- Tell me about a time you experienced conflict with a colleague and how you resolved it.
- Tell me about a time when your project failed. What did you do and what did you learn?
- Do you prefer to work closely with a team or mostly by yourself?
- How are you motivated?

- How do you start learning about a topic you don't know about yet?

23.4.1 Tips for Team Leader Interviews

Our responses help the hiring manager understand:

- How we approach our work.
- The value we bring to the organization.
- Our collaboration and communication style.
- Our professional growth mindset.

During the interview:

- Build rapport with the interviewers.
- Address interviewers by name if we have met them previously.
- Present ourselves as though we are already contributing as a member of the team.
- Discuss the skills we want to develop and the goals we hope to achieve.
- Emphasize both professional growth and our commitment to helping the company succeed.

23.4.2 Unexpected or Creative Questions

Interviewers may ask unexpected questions such as:

"If you could have a superpower, which would it be and why?"

Although these questions may seem unrelated to the role, they often help interviewers evaluate:

- How we respond under unexpected pressure.
- Our creativity.
- Our problem-solving approach.

Answer honestly and enjoy the opportunity to demonstrate our personality.

23.5 Technical Screen

A technical screen is usually conducted by a coding team leader or a senior developer.

Its primary purpose is to evaluate our technical knowledge and determine how well our skills match the requirements of the position.

Most questions focus on technical knowledge and teamwork.

Common questions include:

- What coding languages do you know?
- What are some of your favorite tools?
- How do you begin working on a problem you've been assigned?
- What do you do when you reach something you don't understand?

- How do you stay on schedule and manage your time?
- How do you prepare for a presentation?

Technical interviews also commonly include practical problem-solving exercises.

Interviewers generally do not expect a perfect solution immediately. Instead, these exercises are designed to evaluate:

- Our problem-solving process.
- Our logical thinking.
- Our technical reasoning.
- The strategies we use to approach unfamiliar challenges.

23.6 Conclusion

Regardless of how we feel the second interview went, we should always end the interview on a positive note and clearly express our interest in the position.

Before leaving:

- Thank each interviewer for the opportunity to interview again.
- Express genuine appreciation.
- Offer a sincere smile and handshake.
- Wish the interviewers success in finding the ideal candidate.

Enthusiasm and positivity can leave a lasting impression and help distinguish us from other candidates.

After the interview:

- Ask for the interviewers' contact information if appropriate.
- Promptly send any requested materials, such as links to our portfolio.
- Send thank-you notes on the same day.
- Follow up if we have not received a response within the estimated timeframe.

24 Final Interview

24.1 Learning Objectives

After completing this topic, we will be able to:

- Prepare for a final interview.
- Recognize typical interview questions.
- Explain the **STAR** method for answering interview questions.
- Formulate appropriate questions to ask during an interview.

24.2 Introduction

If we have performed well throughout the interview process, we may be invited for a **final interview**. Traditionally, this interview is conducted live on-site, although many companies now prefer **virtual video interviews**.

The final interview is our opportunity to differentiate ourselves and demonstrate why we are the best fit for the role. Proper preparation helps us present ourselves confidently and professionally.

24.3 Preparing for the Final Interview

24.3.1 Review Before the Interview

On the night before the interview, we should:

- Read through our resume and cover letter.
- Review the job description.
- Research the company.
- Review any notes we have made during the application process.
- Make sure we are familiar with everything mentioned in our resume.
- Rehearse our elevator pitch.
- Practice answering common interview questions.

24.3.2 Plan Our Appearance

We should prepare our appearance in advance so that our grooming and clothing are in excellent condition.

General guideline: Wear professional attire appropriate for the company's work environment.

For men:

- Suit jacket (when appropriate)
- Slacks
- Tie (depending on the workplace)

Many technology companies have a casual office environment, so we should avoid dressing too formally when it is not appropriate.

For women:

- Blouse
- Dress pants
- Business dress

We should avoid wearing anything bright or flashy that could become distracting.

Additional recommendations include:

- Keep hair neat.
- Light makeup is optional.

24.4 Preparing for a Virtual Interview

If our interview is virtual, we should still dress professionally for the camera.

Before the interview:

- Test the camera.
- Test the microphone.
- Ensure the Internet connection is stable.

On the day of the interview:

- Get a good night's sleep.
- Have breakfast.
- Drink coffee if it is part of our routine.
- Exercise if it is part of our routine.
- Make sure we are comfortable and alert.
- Dress and groom ourselves professionally.

24.5 Preparing for an On-Site Interview

If the interview takes place on-site, we should:

- Plan the route to the interview location.
- Arrange transportation.
- Estimate travel time.
- Consider traffic patterns.
- Allow time for parking.
- Account for building check-in procedures.
- Leave early enough to avoid arriving late.

24.6 Understanding Technical Interviews

A technical interview is often very different from other job interviews.

Technical interviews typically include:

- Role-related technical questions
- Behavioral questions
- Situational questions
- Hands-on problem-solving questions
- Trick questions

Interviewers usually begin by asking about our:

- Technical knowledge

- Technical skills
- Professional experience

During these discussions, we should:

- Explain technical concepts in an easy-to-understand manner.
- Describe our work processes clearly.
- Explain our problem-solving process.
- Demonstrate that we understand multiple methods for reaching a solution when more than one approach is possible.

24.7 Situational and Behavioral Questions

Interviewers ask situational and behavioral questions to understand:

- How well we will fit into the project team.
- Whether we align with the company's culture.
- How we approach workplace situations.
- How we solve problems.

The recommended approach for answering these questions is the **STAR method**.

24.8 STAR Method

The **STAR** method helps us explain how we handled specific work challenges in a clear and effective way.

24.8.1 Situation

Describe the project or challenge we faced.

24.8.2 Task

Explain our responsibility in that situation and what we specifically needed to solve.

24.8.3 Action

Describe:

- The steps we followed.
- The procedure we used.
- The decisions we made to solve the problem.

24.8.4 Result

Explain what happened after our actions, preferably using measurable or quantifiable results.

24.9 STAR Method Example

24.9.1 Interview Question

Can you tell us about the most recent project you worked on?

24.9.2 Example Answer

- **Situation:** We were on a team that was tasked with moving a database to a new server.
- **Task:** We were responsible for increasing the query speed of the new deployment.
- **Action:** We employed a data optimizer, improved the indexes, and updated the software.
- **Result:** This produced a **40% increase in query response time**.

24.10 Common Behavioral Interview Questions

Some common behavioral interview questions include:

- What would you do if you were asked to perform a task and were not sure how to complete it?
- Tell me about a time when you went above and beyond your expected tasks while working on a project.
- How would you overcome the challenge of working with a difficult coworker on a team project?

If we need additional information to answer a question properly, we should ask the interviewer for clarification. This demonstrates that we are comfortable asking questions when necessary and want to provide an accurate response.

24.11 Questions to Ask the Interviewer

At the end of the interview, we are usually given an opportunity to ask questions.

This allows us to:

- Demonstrate genuine interest in the role.
- Show that we have researched the company.
- Learn whether the position is the right fit for us.

Some good questions include:

- What is the next step in the interview or hiring process?
- How long does your recruitment process usually take?
- What are the primary responsibilities of this position?
- What would my day-to-day routine look like if I got the job?
- What can you tell me about the job apart from what was included in the description?

24.12 Questions to Avoid

We should avoid asking questions such as:

- How many holidays do you offer?
- Do you have long working hours?
- Do you have another open position?
- What does the company do?

Generally, we should avoid questions that make us appear:

- Uninterested
- Careless
- Ill-informed

It is also better not to ask very specific questions about compensation during the final interview. However, it is usually acceptable to ask about:

- A salary range or pay scale.
- General employee benefits.

If we have reached the final interview stage, the employer is seriously considering us and should be willing to provide this information. However, the final interview is generally **not the appropriate time to negotiate compensation**. We will usually have the opportunity to discuss negotiation if we receive a job offer.

24.13 Key Takeaways

In this topic, we learned that:

- Arriving at an interview well-prepared and presenting ourselves professionally is important.
- There are several types of questions we can expect during a final interview.
- The **STAR** method helps us provide strong, structured answers to situational interview questions.
- Asking thoughtful questions helps us gather valuable information and demonstrate our interest in the role.
- Some interview questions should be avoided because they may create a negative impression.
- Final interviews can be stressful, but with proper preparation, we can present ourselves confidently, demonstrate that we are an excellent fit for the role, and show that we would be a positive addition to the team.

25 After the Interview

Waiting for a response after any job interview can be stressful. After sending a thank-you note to the person or interview panel who took the time to interview us, we should reflect constructively on the experience and consider what would happen next if we receive an offer—or if we do not.

It is useful to track the process by noting:

- The company that interviewed us.
- The interview date.
- The personnel we met.
- The outcome of the interview.
- The questions we were asked and our answers.
- Whether we would answer any questions differently in the future.

We should remember that our job search is not over until we secure a new position, so we need to continue preparing ourselves throughout the process.

25.1 If We Get an Offer

Receiving a job offer is always exciting. However, we should not get swept up in the excitement and forget that we do not have to accept the offer immediately. We should carefully evaluate every aspect of the offer before making a decision.

We can tell the hiring manager that we need some time to review the offer before deciding. The recruiter may suggest a decision timeline and encourage us to ask questions about the offer package if needed.

Generally, we should make our decision within **two business days**, as companies often want to complete the hiring process quickly. Waiting too long may cause the employer to move on to another candidate.

25.1.1 Evaluating Compensation

If the salary or employee benefits do not match what we consider fair, we should:

- Research the current market.
- Present our strengths and selling points.
- Explain the valid reasons why we believe a higher salary is appropriate.

Requesting a fair compensation package is completely acceptable.

When evaluating compensation, we should consider both salary and employee benefits, including:

- Signing bonuses.
- Moving or relocation allowances.
- Stock options.
- Other non-salary compensation.
- Flexible work schedules.
- A shorter or more convenient commute.
- Other valuable workplace benefits.

25.1.2 Negotiating a Salary

Negotiating a salary offer may feel uncomfortable, but if the offer is below our expectations after considering the entire compensation package, we may choose to negotiate.

We can respond by phone or email and should:

- Express excitement and appreciation for receiving the offer.
- Explain how we plan to contribute to achieving the company's goals.
- Base our counteroffer on market research and verifiable evidence of the value we bring to the organization.

Some legitimate arguments that can support salary negotiations include:

- Commute and relocation expenses.
- Having a university degree that was not required for the position but provides academic preparation that can help exceed job performance expectations.
- Possessing specialized skills or certifications that required extensive or costly training, particularly if they were listed as preferred qualifications in the job posting.

Regardless of the company's response, we should always thank them for considering our request. The company may be unable to increase the offer because of budget limitations or company policies regarding salary ranges, but respectfully asking is acceptable.

25.1.3 Withdrawing Other Applications

If we are between interview rounds or waiting for another offer after accepting a position, we should contact the other companies and formally withdraw our applications.

When doing so, we should:

- Thank them for their consideration.
- Inform them that we have accepted another offer.
- Ask them to keep our information on file for future opportunities.
- Remain professional and courteous.

Maintaining a positive relationship is important because we may want to apply there again in the future.

25.1.4 Resigning from a Current Job

If we need to resign from our current position, we should provide the employer with the standard **two weeks' notice** whenever possible.

We should always behave professionally and preserve a good reputation, since we may need a reference in the future or even return to the company in a different role.

If possible, we should arrange our start date with the new employer so that we can take a short break between jobs. Beginning a new position after a few days of rest can make the transition and onboarding process smoother than starting immediately after leaving the previous job.

25.2 If We Are Declined

If we are not selected for a position, the most important step is to learn from the experience.

Every job interview provides valuable lessons and serves as an excellent reference for future job applications.

We should remember that not being selected is not a personal reflection of our abilities. Only one candidate can be hired, even when several applicants are highly qualified. Since we may never know all the factors behind the hiring decision, we should avoid dwelling on the outcome.

25.2.1 Requesting Feedback

We should contact the interviewer by phone or email to:

- Thank them for notifying us of the decision.
- Express appreciation for their time and consideration.
- Request honest feedback about our interview performance.
- Ask for advice on areas where we can improve.

We should keep an open mind, especially if the feedback highlights areas we had not previously considered.

However, we should also understand that many organizations prohibit employees from providing interview feedback because of legal or company policy reasons.

25.2.2 Reflecting and Improving

If we receive feedback, we should review the entire hiring process, including:

- Our application materials.
- How we prepared for the interview.
- The research we conducted about the company.
- Our interview performance.

We should focus on areas we can improve and create a plan for future interviews.

Some questions to consider include:

- Should we spend more time writing stronger interview answers?
- Should we practice more so that we appear more confident and less awkward?
- Would additional training, reading a relevant book, or receiving coaching from a teacher or professor improve our interview performance?

After identifying areas for improvement, we should develop an action plan and put it into practice.

26 Conclusion

Every interview process provides an opportunity to learn, regardless of the outcome.

If we are not selected, we should view the interview and assessment process as valuable experience that helps prepare us for the job we truly want.

If we are selected, it means our preparation and efforts have paid off.

In either case, we gain valuable knowledge and experience that will benefit us in future job searches, whether our next opportunity comes in a few days or several years from now.

27 Expert Viewpoints: Negotiations

27.1 Introduction

Most candidates find it difficult or intimidating to negotiate a job offer. Negotiating an offer starts at the very beginning of the recruitment process, even before we submit an application. By preparing in advance, researching the market, and understanding our priorities, we can negotiate with confidence and increase the chances of receiving an offer that aligns with our needs and career goals.

27.2 Prepare Before Applying

Before applying for a job, we should take some time to identify our **non-negotiables**.

Write down the things that are critical to us. These may include:

- Salary
- Work location
- Working hours
- Flexibility
- Career progression

These do not have to be limited to the examples above. The important point is to identify what matters most to us and determine what we are not willing to negotiate or compromise on.

27.3 Discuss Your Priorities During the Recruitment Process

Once we know our priorities, we should discuss them with:

- The recruiter
- The hiring manager
- The interviewer

These conversations help us determine whether the opportunity is the right fit.

By the time we receive an offer, we should already know the answers to most of these questions. At that stage, the negotiation should focus on understanding how much we are willing to trade off on some of our priorities.

27.4 Present a Strong Case When Negotiating

When speaking with recruiters, we should clearly explain:

- What we are looking for.
- What is important to us.
- Why we are the right candidate.

Although it may not be another interview, we may still need to sell ourselves again.

We should present a compelling case by explaining:

- What we can do.
- What value we can bring to the organization.
- Why we believe we deserve a higher offer.

For example, if we want to negotiate **\$2,000** more than the initial offer, we should confidently explain the reasons behind the request.

We should have conviction, think carefully about what we are going to say, and enter the negotiation prepared with clear reasons that justify why we are worth the amount we are requesting.

27.5 Research the Market Before Negotiating

Before negotiating, we should research:

- The salary range for the position in the market.
- Our current salary.
- The responsibilities and tasks associated with the role.

One suggestion is to negotiate for approximately **10% to 15% above our current salary**, depending on:

- The role.
- The market salary range.
- The responsibilities of the position.

There is no perfect answer for salary negotiation. The appropriate negotiation depends on the specific job opportunity and the type of work involved.

27.6 It Is Completely Acceptable to Negotiate an Offer

Negotiating an offer is completely acceptable.

If the salary offered is lower than expected, we can say something like:

"I would love to take this position, but the salary is a little on the low side. I feel my market value is higher than what is being offered. Could you explain why the offer is below the current market rate?"

This allows us to understand the employer's perspective while professionally expressing our expectations.

27.7 Negotiate Beyond Salary

Salary is not the only aspect that can be negotiated.

Other important factors may include:

- Home office (remote work)
- Workplace flexibility
- Type of work
- Working arrangements

For example, if working from home is important to us, the employer may explain that spending time in the office helps employees network and build relationships through informal discussions, such as conversations around the coffee machine.

In that situation, we can suggest a compromise, such as:

- Coming into the office two or three times per week.
- Scheduling regular coffee meetings with colleagues.
- Creating regular opportunities for collaboration and networking.

The key is to think carefully about what is most important to us and avoid compromising too much, especially if we expect to stay with the company for several years.

Ensuring that the overall working conditions are right benefits both us and the employer.

27.8 Let the Employer Make the First Offer

One important negotiation strategy is to allow the employer to make the first offer.

After receiving the offer, we can then respond with a counteroffer.

Generally speaking, in negotiations, it is often better not to be the first person to suggest a salary figure.

27.9 Do Not Undervalue Yourself

Negotiation can feel confrontational or intimidating, especially if we have little experience with it.

However, we should never undervalue ourselves.

Before negotiating, we should prepare by researching:

- The position we are applying for.
- Typical salaries for similar jobs.
- The level of experience required.
- The technical skills required for the role.

We should conduct online research to determine the typical salary range for the position.

27.10 Determine Your Minimum Acceptable Salary

Before entering a negotiation, we should decide on the **minimum salary** we are willing to accept.

Once we know this number, we should make an initial request that is **slightly higher than our minimum acceptable salary**.

This approach is useful because the employer is likely to respond with a counteroffer, increasing the likelihood of reaching an agreement that still meets our minimum expectations.

28 Action Items for Career Preparation

To prepare for a successful job search and career transition, we should complete the following action items:

- Consider our professional options and decide what type of role to pursue.
- Create our resume.
- Gather and publish our portfolio to GitHub.
- Compose and memorize our elevator pitch.
- Draft our basic cover letter.
- Create or update our LinkedIn profile.
- Line up professional references and get their contact information.
- Compose and practice answers for common interview questions.
- Write questions we would like to ask interviewers.
- Research local meetups and professional networking organizations.
- Ask friends to help us rehearse interviews and critique our materials.